When Firms Lead States: Financial Intermediaries and the Politics of Sovereign Green Bond Adoption

Monica Widmann[†]

Abstract

Green bond markets invert the conventional sequence of debt market development: in most countries, corporate issuance systematically precedes sovereign entry by years. This paper argues that the inversion reflects deliberate market construction by global investment banks. The four banks that authored the Green Bond Principles in 2014 concentrated their early contracting activity on private corporations — using repeated transactions to build the verification practices, pricing conventions, and investor networks that constitute a functioning market — and subsequently served as lead managers when sovereigns entered, transferring accumulated expertise across the public-private divide. This bank-built infrastructure enabled sovereign participation on terms states did not design. Cox proportional hazards models confirm that accumulated corporate green bond activity significantly accelerates sovereign adoption, while government agency issuance delays entry by substituting for direct sovereign participation. Together, these findings identify a diffusion pathway that operates neither through interstate emulation nor impersonal market forces, but through the institutional infrastructure financial intermediaries construct between private and public markets — revealing how privately constructed market conditions shape when and on what terms states engage with global climate finance.

Keywords: *Climate finance, green bonds, financial intermediaries, norm diffusion, sovereign debt, market construction*

Word Count: 11,118

[†]Assistant Professor, Seton Hall University, monica.widmann@shu.edu

"The development of a robust and liquid market for green bonds is an important progression for debt markets...In coauthoring these principles we attempt to help standardize the product and we hope to catalyze investment into environmentally sustainable projects, something to which our firm is very committed."
— Suzanne Buchta
Global Co-head of Green Debt Capital Markets, BofA Merrill

1 Introduction

How do new norms reach sovereign borrowers—and who builds the pathways through which they travel? These questions sit at the heart of international political economy, where scholars have long debated whether states or markets drive the diffusion of governance standards. The conventional answer assigns states a leading role: governments define rules, establish benchmarks, and create the institutional conditions under which private actors subsequently operate (Drezner, 2008, Simmons, Dobbin and Garrett, 2006). A parallel logic governs theories of bond market development, where sovereign issuance anchors pricing, legal infrastructure, and investor confidence that enable private sector participation (Dittmar and Yuan, 2008, Durbin and Ng, 2005, Grundy, van Bekkum and Verwijmeren, 2024). Both frameworks converge on a common prediction: when new financial instruments emerge, sovereign participation should precede and enable private adoption.

Green bond markets — debt instruments that earmark proceeds for environmental projects and commit issuers to reporting on their use — confound this expectation. Across the majority of countries, corporate green bonds systematically precede sovereign issuance—often by several years. France saw corporate green bonds in 2012 but sovereign issuance only in 2017. The United Kingdom had corporate activity from 2014 but waited until 2021. The pattern holds in emerging markets as well: Brazil and India both had corporate issuance from 2015 but sovereign entry only in 2023. This reversal is not an artifact of a few outliers. Among countries that eventually issue sovereign green bonds, the median lag between first corporate and first sovereign issuance is four years.

This inverted sequence reflects deliberate market construction by global investment banks and suggests the need to rethink how authority operates in global governance. The four banks

that drafted the Green Bond Principles (GBP) in 2014—Bank of America Merrill Lynch, Citi, Crédit Agricole CIB, and JPMorgan Chase—simultaneously occupied central positions in global debt underwriting. They strategically selected private corporations as early partners because firms offered greater contractual flexibility and lower political risk than government-linked entities. Through repeated corporate transactions, these banks built the institutional infrastructure that defines a functioning green bond market: standardized documentation, verification protocols, pricing conventions, and a dedicated investor base. As this infrastructure accumulated, it lowered the costs and uncertainty that had previously deterred sovereign entry—but it also established the legitimacy standards, reputational benchmarks, and procedural expectations against which sovereign participation would subsequently be evaluated. States entered a market whose rules they did not write, under conditions they did not design.

This argument speaks to foundational debates in international relations about private authority and the sources of governance in the international system (Cutler, Haufler and Porter, 1999). Existing accounts emphasize either state-led standard-setting, in which governments define rules that private actors subsequently adopt (Drezner, 2008, Farrell and Newman, 2019, Simmons, Dobbin and Garrett, 2006), or diffuse market pressures that emerge from decentralized private coordination (Abbott et al., 2021, Hale, 2020). This article identifies a third pathway: *intermediated market construction*, in which financial intermediaries actively assemble the institutional infrastructure through which new norms travel from private to public actors. Banks are not passive conduits between supply and demand; they are institutional entrepreneurs who select partners, develop governance frameworks, and construct the conditions under which sovereign participation becomes feasible—and constrained. The constitutive authority exercised here differs from the rule-making authority documented in studies of credit rating agencies (Sinclair, 2005) or standard-setting bodies: banks are not merely setting rules but building the market reality within which states must subsequently operate. Credit rating agencies exercise power by evaluating actors within a market

that already exists; the GBP consortium exercised power by constructing the market before sovereign actors had decided to participate—establishing the procedural expectations, reputational benchmarks, and investor conventions against which all subsequent entry would be judged.

The implications for climate governance are direct. Sovereign participation in green bond markets—an increasingly important mechanism for financing national climate commitments—follows private market consolidation rather than preceding it. Governments enter under pre-existing institutional and reputational benchmarks established by private actors, limiting their autonomy to define the terms of their own participation. This reveals an indirect but consequential channel through which market structures condition state behavior in global environmental governance, one that operates neither through formal international institutions nor through coercive interstate pressure, but through the market infrastructure that banks construct through repeated private transactions.

The bank-driven diffusion process operates through two mechanisms. First, institutional intermediation: the banks that developed the GBP simultaneously serve as lead managers for both corporate and sovereign debt issuances, enabling them to transfer procedural knowledge across the public-private divide. Second, infrastructure accumulation: repeated corporate transactions build both technical and normative infrastructure that sovereigns can access at lower marginal cost. Technical infrastructure includes investor familiarity, verification capacity, and pricing conventions; normative infrastructure includes the de facto standards and reputational benchmarks against which sovereign entry will be evaluated. Both accumulate through the same bank-mediated corporate transactions.

These predictions are tested using cross-national green bond data covering 10,627 issuances across 94 countries from 2012–2024. Cox proportional hazards models estimate how the cumulative stock of corporate green bond activity affects the timing of a country's first sovereign issuance. The results support the infrastructure accumulation mechanism: each additional private corporate green bond raises the hazard of sovereign adoption by

approximately 1.6–2.2%, with effects compounding as corporate markets mature. An important substitution dynamic also emerges:: government agency bonds—issued by development banks, state-owned utilities, and other public entities—significantly delay sovereign entry by providing alternative channels for public climate finance. Corporate and agency issuance thus influence sovereign decisions through fundamentally different pathways: the former builds infrastructure that facilitates entry, while the latter substitutes for it.

These findings contribute to three literatures. First, they extend theories of norm diffusion beyond the state-led and market-pressure frameworks that dominate international political economy. By demonstrating that financial intermediaries actively construct the institutional conditions for sovereign adoption, this study identifies a diffusion pathway that operates through the deliberate infrastructure-building activities of banks bridging private and public markets. Second, the analysis challenges the sovereign-first model of bond market development by documenting a sequence in which banks—operating through corporate clients—assemble the institutional preconditions for sovereign participation. This pattern carries practical implications for climate finance strategy: the structure of financial intermediation, not merely capital availability or political will, shapes whether and when sustainable finance norms reach sovereign borrowers. Third, the findings illuminate how market structures condition state autonomy in global climate governance. Sovereign green bond programs, increasingly central to national climate strategies, emerge under privately constructed institutional conditions that constrain how governments can participate and on what terms. Understanding who builds these conditions, and how, is essential to explaining the political economy of the global energy transition.

2 The Green Bond Puzzle

2.1 Sovereign-First Sequencing in Traditional Bond Market

Traditional bond market development follows a well-established sequence where sovereign bonds emerge first, creating the institutional infrastructure, benchmark yields, and investor confidence that enable subsequent corporate participation (Dittmar and Yuan, 2008, Durbin and Ng, 2005, Grundy, van Bekkum and Verwijmeren, 2024). Sovereign bonds serve as risk-free reference points, anchoring pricing for all other domestic debt instruments and providing essential infrastructure for private sector borrowing. Governments often enjoy superior credit quality and market access, allowing them to achieve investment-grade ratings before any domestic corporate issuer, thereby accessing international capital markets before domestic corporate issuers can do so. Additionally, sovereign issuance can create positive spillovers for corporate markets by developing essential market infrastructure, such as primary dealer networks, secondary trading systems, and regulatory frameworks, lowering entry costs for private borrowers.

This sovereign-first model has guided both academic understanding and policy practice in debt market development. International financial institutions routinely advise developing countries to establish sovereign bond programs as the foundation for broader capital market development, and empirical research consistently finds evidence of sovereign leadership in market sequencing (Presbitero et al., 2015). From this perspective, private debt markets are understood to build upon public issuance rather than precede it.

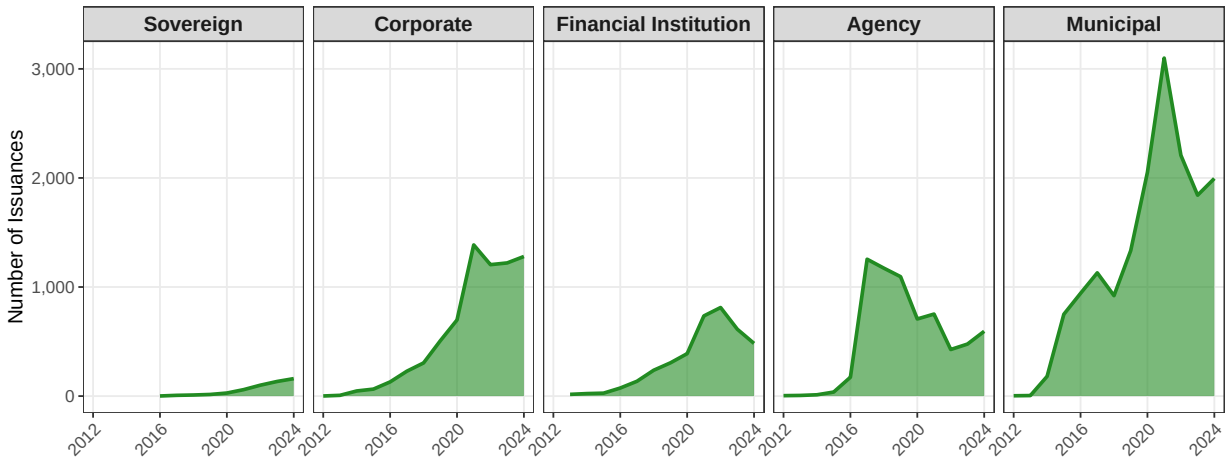
A similar expectation dominates much of international governance. When powerful states agree on regulatory standards, these rules tend to diffuse through formal rule-setting, international institutions, and subsequent adoption by weaker states (Drezner, 2008). In this view, governments act as norm entrepreneurs that define standards ex ante, while private actors adjust their behavior in response to public authority. Applied to financial markets, this framework implies that states should lead in defining credible standards, with private

actors following once rules and benchmarks are established.

2.2 Empirical Reversal in Green Bond Markets

Both expectations—that sovereign bonds anchor market development and that states lead in defining credible standards—suggest that governments should be first movers in green bond markets. Green bond markets,[†] however, reveal a systematic reversal of this conventional order. While the first labeled green bonds were issued by multilateral development banks, beginning with the World Bank in 2008, the subsequent expansion of the market was driven primarily by corporate and financial institution issuers rather than sovereign borrowers. This corporate leadership highlights the private sector–driven nature of green bonds, which contrasts with traditional bond markets.

Figure 1: Number of Green Bond Issuances by Issuer Type



Note: Each panel shows the annual number of green bond issuances for a given issuer type. The y-axis is held constant across panels; the near-flat sovereign panel relative to other issuer types reflects the core empirical pattern motivating this analysis.

Figure 1 illustrates the distinctive issuer composition of green bond markets by number of

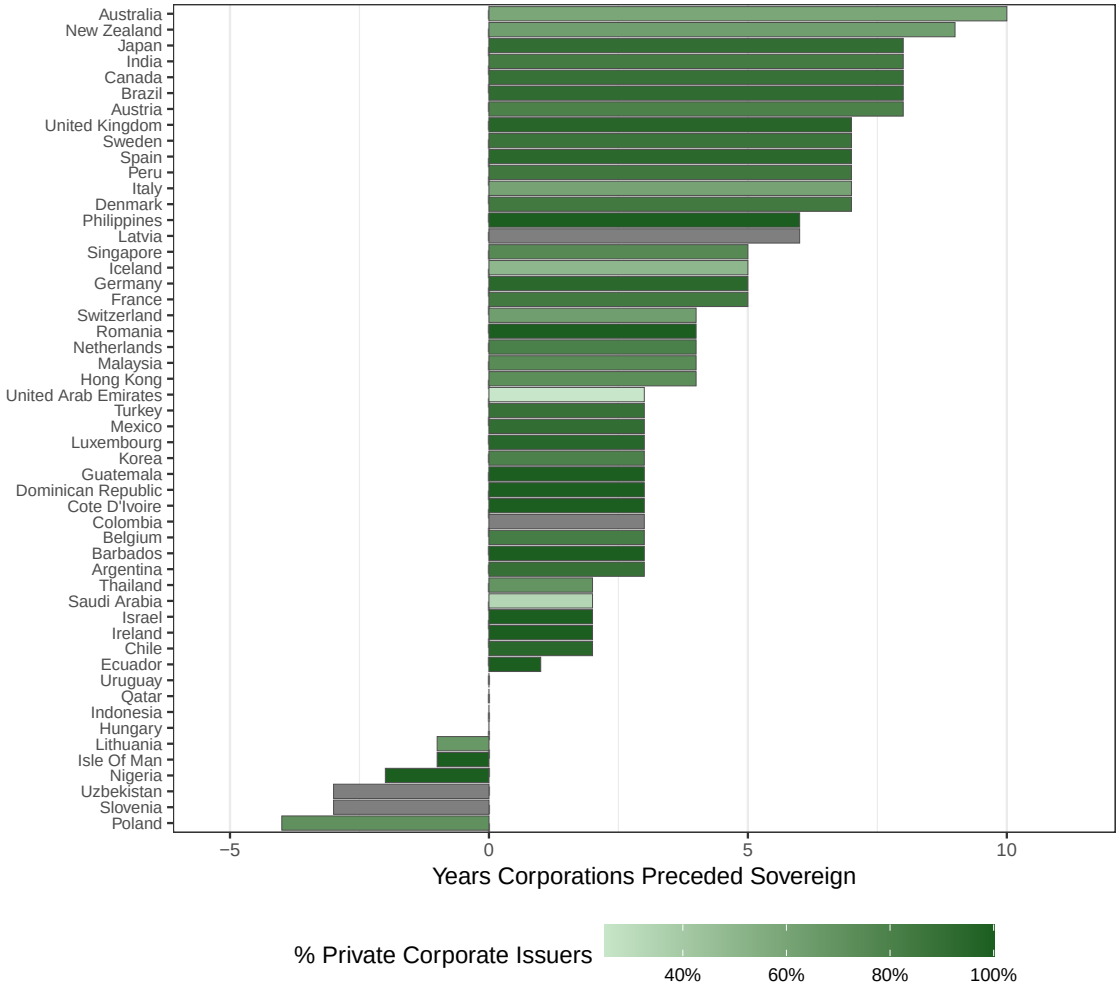
[†]Table A.1 illustrates the range of sustainable bond types. This analysis focuses specifically on green bonds and sustainability bonds, including those funding green initiatives, because they are directly linked to environmental objectives and the diffusion of climate-related norms. Social bonds are excluded, as they primarily finance social projects—such as affordable housing, healthcare, or education—rather than environmental initiatives. The green bond literature has similarly focused on environmentally designated instruments as the relevant unit of analysis (Zerbib, 2025)

issuances. Corporate and financial institution issuers dominate, while municipal issuance — though substantial, driven largely by the United States — reflects subnational infrastructure finance rather than sovereign climate commitment. Agency issuance occupies an intermediate position, financing public priorities through delegated entities rather than direct sovereign programs. Sovereign issuances, by contrast, remain strikingly low throughout the period. This distribution contrasts sharply with the conventional expectation that sovereign issuance anchors bond market development.

Figure 2 provides the clearest evidence of this departure from conventional patterns. The lag analysis, calculated as the difference between the year of first sovereign green bond issuance and the year of first corporate green bond issuance, shows predominantly positive values across countries — indicating that corporate sectors systematically preceded sovereign adoption. Bar shading reflects the share of private (non-government-linked) corporate issuers in each country, with darker green indicating a higher proportion of private market activity. The pattern holds across both advanced economies — Germany (5-year lag), the United Kingdom (7-year lag), and Canada (8-year lag) — and emerging markets — Brazil (8-year lag), India (8-year lag), and the Philippines (6-year lag). Crucially, in most of these cases the bars are dark green, indicating that this corporate leadership was driven primarily by private firms rather than government-linked entities. This suggests that sovereign green bond adoption was not simply a coordinated public sector effort, but rather a response to signals emanating from private capital markets.

Table 1 quantifies this relationship further. Countries with high levels of corporate private green bond activity (10+ bonds) show an 85.0% sovereign adoption rate, compared to only 42.5% among countries with no corporate private green bond activity. This dose-response pattern suggests that corporate activity not only precedes but actively facilitates sovereign market entry. The temporal gaps are substantial and persistent. Among countries that eventually issue sovereign green bonds, the median lag between first corporate and first sovereign issuance is 4 years. These extended delays cannot be explained by short-term

Figure 2: Corporate Leadership in Green Bond Issuance



Note: The bars represent the lag between the first sovereign green bond issuance and the first corporate green bond issuance, calculated as Lag = (year of first sovereign green bond) − (year of first corporate green bond). Positive values indicate corporate-led issuance, where corporate bonds were issued before sovereign bonds in a given country. Negative values indicate cases where sovereign bonds preceded corporate bonds. Bar color reflects the share of private (non-government-linked) corporate issuers in each country, with darker green indicating a higher share of private issuers. The figure demonstrates that, in most countries, corporate green bonds were issued before sovereign green bonds, and that this leadership is predominantly driven by private rather than government-linked corporations.

market conditions or temporary policy constraints—they represent systematic differences in how green and traditional bond markets develop.

Notably, the 42.5% adoption rate among countries with no corporate private activity does not undermine the infrastructure accumulation mechanism. Of the 17 sovereign adopters in this category, 10 had no meaningful green bond ecosystem of any kind—no agency, municipal,

Table 1: Corporate Private Activity and Sovereign Green Bond Market Entry

Corporate Private Green Bond Level	Countries	Sovereign Adopters	Sovereign Adoption Rate (%)
High Activity (10+ bonds)	40	34	85.0
Medium Activity (5–9 bonds)	11	6	54.5
Low Activity (1–4 bonds)	25	9	36.0
No Green Bond Activity	40	17	42.5

Note: Analysis includes all countries in the sample. Corporate private activity is measured as the total cumulative issuance of private corporate green bonds over the sample period. Sovereign adoption rate is the percentage of countries in each activity category that have ever issued a sovereign green bond. Unlike the Cox models, which use time-varying covariates and incorporate the temporal ordering of events, this descriptive table is cross-sectional and does **not** account for *when* corporate private activity occurred relative to sovereign issuance. The lower adoption rate among low-activity countries (1–4 bonds) relative to those with no corporate activity reflects the composition of the no-activity group: the majority of sovereign adopters in that category are small states that issued green bonds as political signaling instruments in conjunction with international climate commitments, rather than in response to accumulated market infrastructure (see text). Countries with nascent but insufficient corporate market development (1–4 bonds) do not benefit from this signaling pathway and have not yet accumulated the infrastructure that drives adoption at higher activity levels, producing the observed non-monotonicity. This pattern is consistent with the infrastructure accumulation mechanism, which predicts meaningful effects at scale rather than a linear response to any level of issuance.

or other non-sovereign issuance. These are predominantly small states (e.g., Fiji, Seychelles, Belize, Benin, Gabon) that issued sovereign green bonds as political signaling instruments, often in conjunction with international climate commitments, rather than in response to accumulated market infrastructure. Eleven of the 17 adopted in 2021 or later, coinciding with the post-Paris cycle and COVID-era green recovery initiatives.

2.3 Why Green Bonds Deviate

Green bonds differ from conventional debt instruments in ways that alter the expected sequence of market development. Unlike traditional bonds, green bonds embed claims about environmental purpose and impact in addition to credit risk. This introduces forms of normative and informational uncertainty, regarding project eligibility, reporting, and credibility, that sovereign issuance alone does not automatically resolve (Flammer, 2021, Tolliver, Keeley and Managi, 2020). The legal and procedural infrastructure underpinning market access is itself a constructed artifact that shapes who can participate and on what terms (Pistor, 2019). Whereas sovereign creditworthiness anchors pricing in conventional markets, credibility in green bond markets depends on demonstrating compliance with evolving environmental norms (Bénabou and Tirole, 2010).

Second, green bond standards emerged through voluntary, market-based coordination rather than formal regulation or intergovernmental agreement. The absence of authoritative

state-defined rules meant that no single government could unilaterally establish credible benchmarks ex ante. Instead, standards evolved through repeated transactions and market acceptance, favoring actors capable of coordination (Abbott et al., 2021, Hale, 2020).

Third, sovereign green bond issuance carries political and reputational stakes that differ from those associated with conventional borrowing. Issuing a green bond constitutes a public commitment to environmental objectives and exposes governments to accusations of greenwashing or policy inconsistency if proceeds are contested (Flammer, 2021). These risks create incentives for governments to delay entry until market conventions and investor expectations are sufficiently stabilized.

These features help explain why green bond markets do not follow the sovereign-first sequencing observed in conventional debt markets. They also create conditions under which private actors can play a leading role in early market development—setting the stage for the bank-mediated diffusion processes examined in the next section.

3 Banks as Institutional Entrepreneurs in Green Finance

The conventional account of green bonds emphasizes investor demand for sustainable assets and issuers' incentives to signal environmental commitment and credibility (Baker et al., 2021, Flammer, 2021, Riedl and Smeets, 2017, Tang and Zhang, 2020, Zerbib, 2025).[‡] This supply-and-demand framing, however, obscures the pivotal role of a third set of actors: the global investment banks that created the market's institutional infrastructure. Banks functioned as institutional entrepreneurs that developed the Green Bond Principles (GBP) and actively diffused green bonds through corporate client relationships, building the market conditions that eventually enabled—and encouraged—sovereign participation. This authority is constitutive rather than evaluative: unlike credit rating agencies, which assess actors within

[‡]The concept of institutional entrepreneurship originates with DiMaggio (1988), who defines institutional entrepreneurs as actors who leverage resources to create or transform institutions in ways that serve their interests.

a market that already exists, the GBP consortium constructed the market itself—its procedural expectations, reputational benchmarks, and investor conventions—before sovereign actors had decided to participate.

3.1 Why Banks Created the Green Bond Principles

The green bond market did not emerge organically from dispersed market transactions. It was deliberately constructed by a consortium of investment banks seeking to transform a one-off product into a scalable asset class. The first green bonds, issued by the European Investment Bank (2007) and World Bank (2008), were structured as one-off transactions between multilateral development banks and institutional investors. These early deals demonstrated proof of concept but remained confined to AAA-rated supranational issuers. By 2013, green bonds were attracting interest from corporate issuers and a broader investor base, but without standardized processes for designating, verifying, and reporting on green bond proceeds, each transaction required individual negotiation.

The solution emerged from informal coordination among debt capital markets professionals at the 2012 Environmental Finance conference. Michael Eckhart, then Global Head of Environmental Finance at Citigroup, later explained why bankers—rather than environmental groups or regulators—took the lead: "We agreed... that we're going to have to write this guideline for so-called green bonds because nobody else except us seemed to understand the bond market" (Eckhart, 2021).[§] The resulting GBP, announced in January 2014, was drafted by four banks—Bank of America Merrill Lynch, Citi, Crédit Agricole CIB, and JPMorgan Chase—with nine additional global banks joining as initial signatories (Bank of America Merrill Lynch et al., 2014).

The document was deliberately minimalist: four pages specifying process requirements for use of proceeds, project evaluation, management of proceeds, and reporting—without defining what counted as "green." This design choice was strategic. As Eckhart explained:

[§]See Appendix H for additional evidence on bank motivations.

"The bond market doesn't need a definition of green. It needs an *opinion* [that] something is green" (Eckhart, 2021). By focusing on process rather than substance, the GBP created a framework flexible enough to accommodate diverse issuers while providing sufficient standardization to facilitate transactions.

Three interrelated motivations drove banks to invest in this institutional infrastructure. First, standardization was necessary to *scale* the market beyond individual MDB transactions. Without common documentation and processes, transaction costs remained prohibitively high for smaller issuances. Second, banks sought to establish *credibility* in a market vulnerable to greenwashing accusations. Having witnessed how reputational damage destroyed structured credit products in 2008, the drafting banks understood that voluntary standards could preempt similar risks (Ceci, 2024). Third, banks had *commercial interests* in building a market they could intermediate. Green bond underwriting generates fees comparable to conventional bond issuance, and the banks that established market standards positioned themselves as natural counterparties for new issuers.

3.2 From Corporate Clients to Sovereign Adoption

The creation of the GBP was necessary but not sufficient for market development. The same banks that drafted the principles also occupied central positions in global debt underwriting, enabling them to actively promote green bonds to their corporate clients. Investment banks maintain ongoing relationships with corporate treasurers and CFOs, advising on capital structure decisions and executing debt issuances. When a bank underwrites a green bond, it provides not just capital market access but also the procedural knowledge necessary for GBP-compliant issuance: developing a green bond framework, engaging external reviewers, establishing tracking systems for proceeds, and preparing impact reports.

Crucially, founding banks concentrated their early contracting activity on private corporations rather than government-linked entities. This pattern reflects strategic selection under political constraint: private firms offered greater contractual flexibility and lower po-

litical risk than public entities, making them ideal partners for establishing and refining the procedural standards that would govern the market. Private corporate issuers thus served as governance laboratories during the market's formative phase. Through repeated private corporate transactions, banks accumulated institutional infrastructure, such as standardized documentation, trained intermediaries, established verification providers, and familiar investor bases, that reduced uncertainty and lowered the marginal cost of subsequent sovereign entry.

Private corporate activity influences sovereign adoption through two related pathways. First, *institutional intermediation*: the same banks that underwrite corporate green bonds also serve as lead managers for sovereign debt issuances, transferring procedural knowledge from corporate deal experience to sovereign clients. Second, *infrastructure accumulation*: repeated corporate issuance builds both the technical and normative infrastructure that sovereigns can access at lower marginal cost. On the technical side, this includes the investor base, verification capacity, and pricing conventions that constitute a functioning market. On the normative side, corporate adoption of GBP-compliant frameworks creates the reputational benchmarks and expectations against which sovereign entry will be evaluated—a sovereign green bond will be assessed against standards sediment through prior corporate transactions.

Private corporate green bond issuance therefore influences sovereign adoption not primarily through imitation or signaling, but through infrastructure accumulation. Each corporate issuance expands the stock of market institutions on which sovereigns can later draw. This logic yields the central empirical implication of the argument:

> **Hypothesis 1:** *Higher levels of prior private corporate green bond issuance increase the likelihood and accelerate the timing of a country's first sovereign green bond issuance.*

3.3 Agency Bonds as Substitutes

Not all non-sovereign green bond issuance operates through diffusion. Government agency bonds—issued by development banks, state-owned enterprises, public utilities, and other government-sponsored entities—occupy an institutional space between corporate and sovereign debt. These are public sector entities with specific mandates but legal and financial autonomy from the central government. For example, the *Tennessee Valley Authority* in the United States is a federally owned corporation that issues its own bonds to finance electricity generation and infrastructure projects, including green bonds for renewable energy initiatives. While such agency bonds are not direct obligations of the central government like sovereign bonds, they typically carry implicit government backing and are perceived by markets as quasi-sovereign instruments. Although legally distinct from sovereign bonds, agency bonds often carry implicit state backing, finance similar categories of infrastructure, and draw on overlapping investor bases. As a result, agency bonds function as close *substitutes* for sovereign issuance rather than as complements.

Government agencies are natural early participants in green bond markets for reasons that distinguish them from both sovereigns and private corporations. Development banks, state-owned utilities, and public infrastructure authorities often have dedicated environmental mandates—financing renewable energy, public transit, or water infrastructure—that provide clearly defined project pipelines aligned with green bond eligibility criteria. This narrow operational focus reduces the verification burden and greenwashing risk that complicate sovereign issuance, where proceeds must be allocated across a broader and more politically contested set of priorities. Agencies also enjoy greater operational autonomy from electoral cycles, enabling faster issuance decisions without the political deliberation that accompanies sovereign debt programs. Several of the most active agency issuers reflect these dynamics: the United States, where federally sponsored entities like the Tennessee Valley Authority and Federal Home Loan Banks account for over 9,000 agency green bonds; Japan, where public development institutions issued 324 agency bonds before the sovereign entered in 2024; and

Norway, where state energy and infrastructure entities have issued 48 agency bonds without any sovereign issuance to date (Table 9).

From the perspective of central governments, agency issuance provides an alternative pathway for financing climate-related investments without requiring the political commitment and reputational exposure associated with direct sovereign green bonds. When agencies already supply substantial green financing capacity, governments face reduced fiscal necessity and weaker incentives to enter the sovereign green bond market. Unlike corporate issuance, which expands market scope and infrastructure, agency issuance replicates core sovereign functions through delegated entities.

The substitution effect operates through two reinforcing mechanisms. First, agency issuance directly satisfies climate finance needs, reducing the marginal benefit of sovereign participation. When major environmental investment priorities are already funded via agency bonds, central government issuance may appear redundant from both budgetary and policy perspectives. Second, sustained and well-received agency issuance creates an infrastructure accumulation that differs from corporate demonstration. When agencies consistently issue green bonds, governments may conclude that the symbolic, political, or market-building rationale for sovereign entry is diminished.

In contrast, corporate green bond issuance is structurally and institutionally distinct from sovereign debt. Corporate issuers bear private sector credit risk, appeal to partially different investor pools, and finance projects, such as energy efficiency retrofits, sustainable supply chain innovations, and green technology R&D, that governments typically do not directly fund. This structural difference makes corporate issuance complementary to rather than substitutive for sovereign green bonds.

> **Hypothesis 2:** *High levels of green bond issuance by government agencies reduce the likelihood of sovereign green bond adoption by meeting government financing needs through alternative channels.*

4 Data

4.1 Green Bond Issuance Data

This paper draws on a dataset of green bond issuances spanning 2012–2024, recording bond-level information including issuer name, issuer type, country of issuance, settlement date, issuance value, and lead managers. The analysis captures self-labeled green bonds — issuances in which the issuer declares the bond green and commits to the GBP's process requirements, typically with an external second-party opinion, but without independent certification that the underlying projects meet a defined environmental standard. This is the appropriate population for the theoretical argument advanced here. The claim is not that green bonds fund genuinely green projects, but that the GBP framework — its procedural requirements, verification conventions, and documentation standards — constitutes the market infrastructure through which banks enabled sovereign adoption. That infrastructure was built through the self-labeled market, which is the universe within which the GBP operates.

An alternative approach would restrict analysis to bonds certified by the Climate Bonds Initiative (CBI), which maintains its own project taxonomy and independently verifies that proceeds fund environmentally credible investments. CBI certification represents a stricter standard, but covers a substantially smaller share of issuances and is systematically absent from the early-period activity — 2012 through roughly 2016 — that is central to the infrastructure accumulation argument. Restricting to CBI-certified bonds would therefore introduce selection bias precisely where the theoretical claims are most consequential, undersampling the founding bank activity documented in Section 5 and attenuating the infrastructure effects estimated in Section 7.

After excluding supranational issuers, the dataset covers 10,627 green bond issuances across 94 countries, including 6,118 agency bonds, 5,798 corporate bonds, and 354 sovereign bonds. Sovereign bonds account for the smallest share of issuances but the largest average transaction size ($1.32 billion), consistent with the pattern in conventional debt markets

where sovereign deals are fewer but substantially larger than corporate offerings. Corporate bonds represent the largest total issuance value ($1.38 trillion), reflecting the high volume of private sector activity that is central to the analysis.

4.2 Issuer Classification

Issuers are classified into five broad categories based on their institutional characteristics and relationship to government: sovereign, agency, municipal, corporate, and financial institution. For corporate and financial institution issuers, the analysis further distinguishes between *private* and *government-linked* entities, yielding seven analytical categories. Government-linked issuers include state-owned enterprises, government-sponsored entities, and firms with substantial government ownership or control. Classification was conducted manually using ownership information, regulatory filings, and public records.

The distinction between private and government-linked issuers is central to the analysis. Private corporations operate under different political constraints and offer greater contractual flexibility than government-linked entities. If banks strategically selected contracting partners during market formation, differential patterns of bank involvement should be observable across these issuer types.

4.3 Lead Manager Identification

For each bond issuance, the lead managers responsible for underwriting and distributing the bond are identified. The lead manager field is parsed to extract individual banks, and names are standardized across issuances. Lead managers are then classified according to their role in creating the Green Bond Principles (GBP).

Founding banks are the four institutions that drafted the GBP in January 2014: Bank of America Merrill Lynch, Citi, Crédit Agricole CIB, and JPMorgan Chase. *Initial supporters* are the nine banks that joined as signatories at the GBP launch: BNP Paribas, Daiwa Securities Group, Deutsche Bank, Goldman Sachs, HSBC, Mizuho Financial Group, Morgan

Stanley, Rabobank, and SEB. *Early adopters* are twelve additional banks that joined shortly after launch, including Barclays, Credit Suisse, and Société Générale. Together, these 25 banks constitute the GBP consortium.[¶]

This classification enables an assessment of whether the banks that created green bond standards also dominated early market activity, and whether their involvement varied systematically across issuer types and over time.

5 Bank Leadership in Market Formation

The theory predicts that the banks which created the GBP deliberately concentrated their early contracting activity on private corporations to establish market standards through flexible, iterative deal structures. This section tests three empirical implications of the infrastructure accumulation framework: (1) founding banks should dominate market activity *before* the GBP's formal launch; (2) founding bank involvement should be systematically higher with private corporates than government-linked entities; and (3) founding bank participation should decline after 2014 as accumulated infrastructure reduced the need for direct involvement.

5.1 Pre-Institutional Market Dominance

The analysis begins by examining the composition of green bond issuance in 2012–2013, prior to the GBP's January 2014 launch. If banks were responding to an already existing market rather than constructing one, diverse underwriting patterns with no systematic concentration among the institutions that later codified standards would be expected. Instead, Table 2 reveals striking concentration.

[¶]The 25 GBP consortium banks are as follows. Founding banks (4): Bank of America Merrill Lynch, Citi, Crédit Agricole CIB, and JPMorgan Chase. Initial supporters (9): BNP Paribas, Daiwa Capital Markets America, Deutsche Bank, Goldman Sachs, HSBC, Mizuho Securities, Morgan Stanley, Rabobank, and SEB. Early adopters (12): Barclays, BMO Financial Group, Credit Suisse, DNB, DZ Bank, ING, Lloyds Bank, Mitsubishi UFJ Securities, Nomura, RBC Capital Markets, Santander Global Banking & Markets, and

Table 2: Pre-GBP Green Bond Market Composition (2012–2013)

GBP Consortium Role	Bonds	Countries	Managers	Share (%)
Founding Bank	12	6	4	92.3
Initial Supporter	6	5	5	46.2
Additional Supporter	2	2	2	15.4
Any GBP Consortium Bank	13	7	11	100.0
Non-Consortium Bank	0	0	3	0.0

Note: Analysis covers all green bond issuances from January 2012 through December 2013, prior to the Green Bond Principles' January 2014 launch. Founding Banks are the four institutions that drafted the GBP (Bank of America Merrill Lynch, Citi, Crédit Agricole CIB, JPMorgan Chase). Initial Supporters are the nine banks that joined as signatories at launch. Additional Supporters joined within six months of launch. Share is calculated as the percentage of bonds with at least one lead manager from each category. Categories are not mutually exclusive; a single bond may have multiple lead managers from different consortium roles.

Founding banks—the four institutions that drafted the GBP—underwrote 92.3% of all green bonds issued before the principles existed. Initial supporters, the nine banks that joined at launch, underwrote 46.2%. Together, GBP consortium banks captured 100% of the pre-GBP market. This complete market coverage occurred despite the absence of formalized standards, investor mandates, or regulatory requirements for green bond issuance.

The pattern suggests these banks were not intermediating a pre-existing demand for standardized green bonds, but rather creating the market conditions that would make standardization valuable. The 13 bonds in this period, spanning 7 countries and involving 11 different lead managers, represented a coordinated effort to establish proof of concept across jurisdictions and issuer types. Notably, no bonds in this period were underwritten by institutions outside the eventual GBP consortium, indicating that early market participation was restricted to the banks that would subsequently formalize the standards.

Table 3 identifies the specific banks that dominated this formative period. Crédit Agricole CIB led with 9 bond issuances across 3 countries, followed by Bank of America Merrill Lynch and SEB with 3 issuances each. These early leaders—all eventual GBP signatories—established the procedural templates, documentation standards, and external verification practices that the GBP would later codify.

Société Générale.

Table 3: Pre-GBP Market Leadership by Bank

Rank	Bank	Bonds	Countries	GBP Role
1	Crédit Agricole CIB	9	3	Founding Bank
2	Bank of America	3	3	Founding Bank
2	SEB	3	3	Initial Supporter
4	BNP Paribas	1	1	Initial Supporter
4	Citigroup	1	1	Founding Bank
4	DZ Bank	1	1	Additional Supporter
4	HSBC	1	1	Initial Supporter
4	JPMorgan	1	1	Founding Bank
4	Morgan Stanley	1	1	Initial Supporter
4	Rabobank	1	1	Initial Supporter

Note: Analysis covers 2012–2024 issuances, excluding municipal, agency, and sovereign issuers. Government-linked entities include state-owned enterprises, government-sponsored entities, and firms with majority government ownership. Consortium banks include all 25 GBP founding, initial supporter, and additional supporter institutions. Founding banks are the four GBP drafters. Share is calculated as the percentage of bonds in each category with at least one consortium or founding bank as lead manager. The table excludes 3,008 municipal bonds and 181 agency bonds for clarity.

This temporal sequence contradicts a simple demand-side explanation. If investor appetite for green bonds had driven market development, diverse financial intermediaries would be expected to compete for underwriting mandates. Instead, a small consortium of banks dominated early transactions and subsequently authored the standards that defined the market. This pattern indicates that banks were not merely responding to existing demand but constructing the institutional infrastructure—deal precedents, verification protocols, and pricing conventions—necessary for market scaling.

5.2 Strategic Selection of Private Corporate Partners

The theory further predicts that founding banks should concentrate their contracting activity on private corporations, which offer greater flexibility for establishing governance norms than government-linked entities constrained by public accountability and regulatory oversight. Table 4 evaluates this prediction by disaggregating bank involvement across issuer types.

The evidence supports strategic partner selection. Founding banks participated in 21.6% of private corporate issuances but only 13.1% of government-linked corporate issuances—a 65% difference in involvement rates. This gap is substantial: despite government-linked

Table 4: Bank Involvement by Issuer Sector and Ownership Structure

Sector	Ownership	Bonds	Consortium Bonds	Founder Bonds	Consortium Share (%)	Founder Share (%)	Countries
Corporate	Private	464	302	100	65.1	21.6	26
Corporate	Government-linked	214	126	28	58.9	13.1	13
Financial	Private	106	75	35	70.8	33.0	14
Financial	Government-linked	279	150	65	53.8	23.3	14

Note: Analysis covers 2012–2024 issuances, excluding municipal, agency, and sovereign issuers. Government-linked entities include state-owned enterprises, government-sponsored entities, and firms with majority government ownership. Consortium banks include all 25 GBP founding, initial supporter, and additional supporter institutions. Founding banks are the four GBP drafters. Share is calculated as the percentage of bonds in each category with at least one consortium or founding bank as lead manager. The table excludes 3,008 municipal bonds and 181 agency bonds for clarity.

corporates issuing 214 bonds compared to 464 private corporate bonds, founding banks systematically prioritized private entities even though government-linked issuers often carried stronger credit ratings and larger deal sizes.

The pattern extends beyond corporates. Among financial institutions, founding banks participated in 33.0% of private issuances versus 23.3% of government-linked issuances. This cross-sectoral consistency indicates that ownership structure, rather than sector-specific factors, drove bank selection. Private issuers—whether corporate or financial—provided the contractual flexibility necessary for iterative standard-setting without the political constraints and disclosure requirements that govern state-linked entities.

The contrast with broader consortium involvement is instructive. While founding banks showed differentiation (21.6% vs. 13.1%), the full GBP consortium exhibited more uniform participation rates (65.1% for private corporates vs. 58.9% for government-linked). This suggests that founding banks, as standard-setters, required specific issuer characteristics during the governance formation phase, while later consortium members could apply established templates more uniformly.

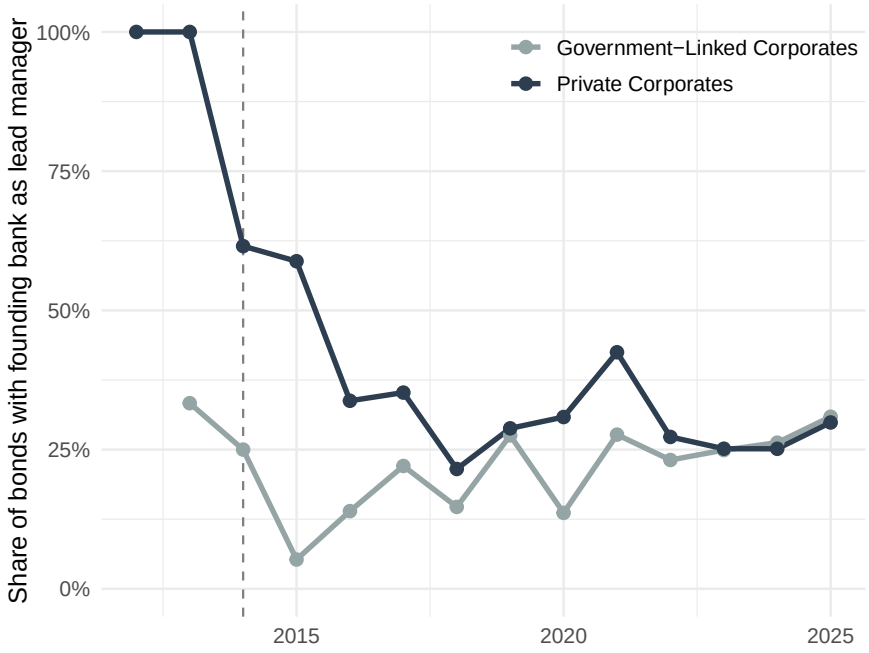
The relatively small volume of private corporate issuance might appear to undercut its theoretical importance. Private corporates represent only 464 of 4,252 total bonds in the sample (10.9%). Yet these bonds account for 75 unique issuers across 26 countries. This geographic and issuer diversity, combined with high founding bank involvement, indicates that banks used private corporate deals to establish market precedents across jurisdictions

rather than concentrating on high-volume issuance. The pattern is consistent with a market-building strategy: founding banks prioritized breadth of coverage over depth of activity, seeding green bond infrastructure in diverse regulatory and financial environments where it could subsequently facilitate sovereign entry.

5.3 From Direct Leadership to Market Infrastructure

If founding banks built market infrastructure through concentrated early contracting, their direct involvement should decline once standards stabilized and became self-enforcing. Figure 3 traces founding bank participation over time, comparing private corporates with government-linked corporates to isolate the effect of ownership structure while holding sector constant.

Figure 3: Founding Bank Involvement Over Time: Private vs. Government-Linked Corporates



Note: Figure shows the share of bonds with at least one founding bank as lead manager, by year and corporate ownership type. The dashed vertical line marks the January 2014 GBP launch. Private corporates exhibit complete founding bank coverage (100%) prior to 2014, followed by a sharp decline to approximately 30% by 2017. Government-linked corporates show lower initial founding bank involvement (33%) and a more gradual decline. The two series converge at approximately 25–30% by 2020, indicating that founding banks' concentrated involvement with private corporates was temporary rather than persistent.

Before the GBP launch, founding banks participated in 100% of private corporate green bond issuances, compared to only 33% of government-linked corporate issuances. This three-fold difference during the standard-setting phase (2012–2013) reflects founding banks' strategic selection of partners with maximal contractual flexibility. Private firms, unconstrained by public accountability requirements or political oversight, provided the iterative deal structure necessary to develop credible governance frameworks.

Following the January 2014 GBP release, founding bank involvement with private corporates declined precipitously—falling to 60% in 2015, 30% by 2017, and stabilizing around 25% through 2020. Government-linked corporate involvement followed a more gradual trajectory, declining from 33% to approximately 5% in 2015 before rising to converge with private corporates at 25–30% by 2020. This convergence is theoretically significant: by 2020, founding banks participated in private and government-linked corporate deals at nearly identical rates, suggesting that whatever special role they played in private corporate markets had been exhausted.

This temporal pattern distinguishes the infrastructure accumulation mechanism from persistent market power. If founding banks enjoyed durable competitive advantages from early entry—superior deal expertise, privileged issuer relationships, or proprietary capabilities—their market share should remain elevated or decline gradually. Instead, the abrupt post-2014 decline and subsequent convergence suggest that their initial dominance reflected a functional role in building market infrastructure rather than sustained competitive advantage. Once the GBP provided standardized documentation, established verification procedures, and created investor familiarity, other financial institutions could underwrite green bonds for private corporates without founding bank involvement.

The convergence between private and government-linked corporate series reinforces this interpretation. If founding banks had developed issuer-type-specific expertise through an initial focus on private corporations, persistent differentiation would be expected. Instead, by 2020 both ownership types exhibit statistically indistinguishable founding bank partici-

pation rates. This pattern suggests that the institutional infrastructure created by founding banks—documentation templates, verification protocols, and pricing conventions—becam publicly available and applicable across ownership structures once formalized in the GBP.

Importantly, the decline in founding bank involvement coincided with dramatic market expansion. Annual green bond issuance grew from under 20 bonds in 2013 to over 800 bonds by 2020. This inverse relationship—declining founding bank market share amid rising issuance volume—indicates that their early monopoly did not translate into sustained market control but rather facilitated market development that other intermediaries could subsequently support. The infrastructure founding banks built through concentrated private corporate contracting became the foundation for broader market participation across all issuer types.

5.4 Bank Involvement in Sovereign Market Entry

The preceding analysis documented how founding banks constructed market infrastructure through private corporate clients—using firms as vehicles for developing and testing governance frameworks under conditions of contractual flexibility. Once established, this infrastructure became publicly available. How, then, do sovereigns access it? One possibility is passive diffusion: governments observe established practices and replicate them independently. An alternative is active intermediation: the same banks that built infrastructure through corporate transactions subsequently facilitate sovereign entry by transferring accumulated expertise. If this institutional intermediation pathway operates as theorized, GBP consortium banks should be heavily represented in countries' first sovereign green bond issuances—the critical entry point at which accumulated infrastructure reduces sovereign uncertainty.

5.4.1 Founding Banks as Sovereign Market Gatekeepers

To evaluate this prediction, a deal–manager panel is constructed linking each sovereign green bond issuance to its lead underwriters. Table 5 reports results for country entry deals—defined as the first sovereign green bond issued by each country. The findings support the institutional intermediation mechanism. Of the 59 country entry deals in the sample, 72.9% include at least one GBP founding bank as lead manager. Expanding the definition to include all GBP consortium banks (founders, initial supporters, and early adopters) raises coverage to 93.2%. This near-universal presence of GBP banks at sovereign market entry suggests that the same institutions that built green bond infrastructure through corporate transactions subsequently facilitated sovereign participation.

Table 5: GBP Bank Involvement in Country Entry Deals

Statistic	Value
Total Country Entry Deals	59
Entry Deals with GBP Founder	43 (72.9%)
Entry Deals with Any GBP Bank	55 (93.2%)

Note: Entry deals are defined as the first sovereign green bond issuance by each country. GBP Founders are the four banks that drafted the GBP (Bank of America Merrill Lynch, Citi, Crédit Agricole CIB, JP-Morgan Chase). GBP Banks include all founding, initial supporter, and early adopter institutions.

This pattern is consistent with the transmission mechanism outlined in Section 3: banks that developed procedural templates through corporate contracting subsequently deployed this expertise to facilitate sovereign entry. The near-complete coverage by GBP banks suggests that sovereign issuers rely heavily on intermediaries with established green bond capabilities, rather than developing independent market access.

5.4.2 Market Concentration Over Time

If the infrastructure accumulation mechanism operates as theorized, declining market concentration should accompany the expansion of available templates, trained intermediaries,

and investor familiarity. Early sovereign issuances would be expected to exhibit high concentration among a small number of experienced underwriters, whereas later issuances should display broader participation as green bond capabilities diffuse across the financial sector.

Table 6 and Figure 4 report the Herfindahl-Hirschman Index (HHI) for lead manager positions in sovereign green bonds over time. The pattern strongly supports the infrastructure diffusion prediction. In 2016, when the sovereign market was nascent with only 3 manager slots across all deals, the HHI stood at 0.333—indicating high concentration. By 2022–2023, as the market matured to 321–480 annual manager slots, the HHI declined to 0.037–0.038, reflecting a competitive market with 62–68 distinct underwriters.

Table 6: Lead Manager Concentration in Sovereign Green Bonds

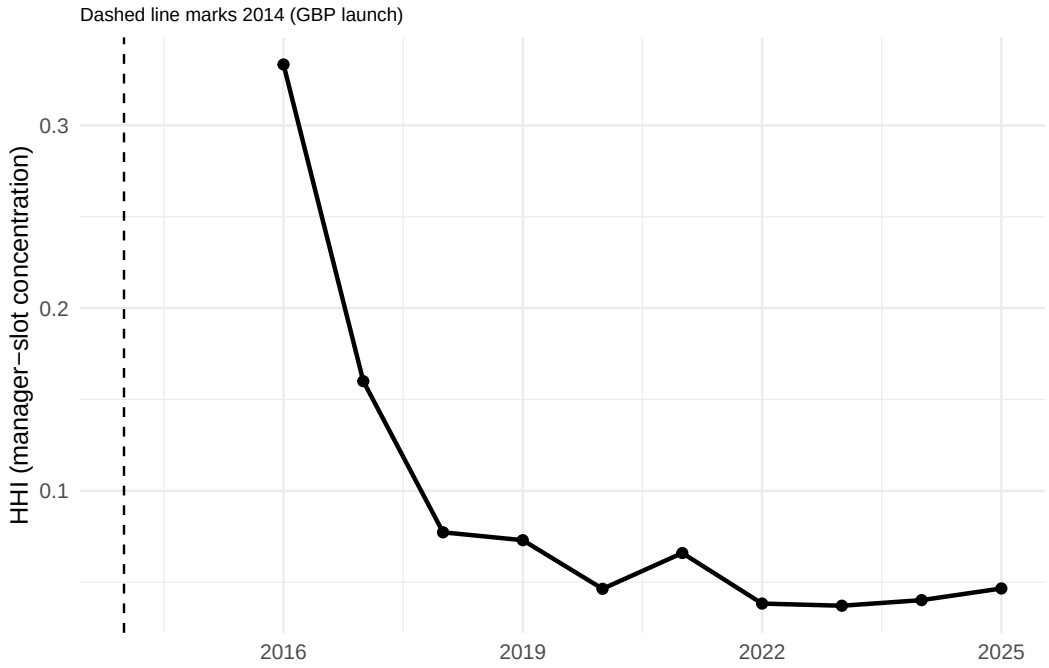
Year	Manager Slots	Distinct Managers	HHI
2016	3	3	0.333
2017	10	8	0.160
2018	29	17	0.077
2019	39	20	0.073
2020	73	33	0.046
2021	248	42	0.066
2022	321	62	0.038
2023	480	68	0.037
2024	278	55	0.040
2025	185	51	0.047

Note: HHI calculated based on manager-slot share. Manager slots represent the total number of lead manager positions across all sovereign green bond deals in each year; a single deal may have multiple lead managers. HHI values above 0.25 indicate high concentration; values below 0.15 indicate a competitive market.

The temporal pattern parallels that observed in corporate markets (Figure 3): founding banks dominated early transactions and then stepped back as accumulated infrastructure enabled broader participation. In the sovereign context, this diffusion reflects both the transfer of green bond expertise across the financial sector and the declining informational advantage of early movers as market practices became standardized.

Notably, the HHI exhibits a modest uptick in 2021 (0.066) before resuming its decline.

Figure 4: Lead Manager Concentration in Sovereign Green Bonds Over Time



Note: Figure plots the Herfindahl-Hirschman Index (HHI) for lead manager positions in sovereign green bonds by year. The dashed vertical line marks 2014, the year of the GBP launch. Lower HHI values indicate greater competition among underwriters.

This likely reflects the surge in sovereign issuance following pandemic-related fiscal expansion, when established underwriters may have captured disproportionate deal flow due to capacity constraints among newer market entrants. The subsequent return to low concentration levels (0.037–0.047) suggests this was a temporary deviation rather than a reversal of the diffusion trend.

6 Empirical Strategy

The preceding analysis documented how founding banks constructed green bond market infrastructure through concentrated contracting with private corporate clients. The next step evaluates whether this bank-built infrastructure systematically accelerates sovereign green bond adoption. Specifically, the analysis examines three empirical implications derived from the theoretical framework: (1) corporate precedence in market entry, (2) cumulative effects of corporate issuance on sovereign adoption, and (3) agency substitution dynamics.

6.1 Model Specification

To examine how corporate green bond market development influences the timing of sovereign adoption, a survival analysis framework is employed using Cox proportional hazards models (Box-Steffensmeier and Jones, 2004). This approach is well suited for analyzing the timing of discrete events—here, a country's first sovereign green bond issuance—while accounting for countries that do not issue during the observation period (right-censoring).

The Cox proportional hazards model is specified as:

$$h(t|X) = h_0(t) \times \exp(\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k) \tag{1}$$

where $h(t \mid X)$ represents the hazard rate of first sovereign green bond issuance at time t for a country with covariates X, and $h_0(t)$ is the baseline hazard function. Hazard ratios above one indicate that a covariate increases the likelihood of sovereign adoption; ratios

below one indicate decreased likelihood. Because the key independent variables measure cumulative issuance, hazard ratios represent the effect of each additional bond added to the accumulated stock, with impacts compounding as issuance accumulates.

The identification strategy leverages the temporal sequence documented in Section 5: founding banks first built market infrastructure through private corporate deals, then formalized this infrastructure in the GBP, creating conditions that facilitated broader market participation. Cumulative corporate issuance therefore serves as a proxy for the stock of available market infrastructure—documentation templates, verification protocols, trained intermediaries, and familiar investor bases—that sovereigns can access when considering market entry.

To account for systematic differences in baseline hazard rates between developed and developing economies, the models are stratified by OECD membership status.[‖] This specification allows OECD and non-OECD countries to have distinct baseline hazard functions while maintaining common covariate effects across groups. Standard errors are clustered at the country level to account for within-country correlation over time. As a robustness check, all models are re-estimated with time-varying coefficients that allow the effects of cumulative issuance variables to evolve as markets mature (Appendix B.2).

The identification strategy relies on the assumption that, conditional on the control variables, variation in the cumulative stock of green bonds from different types of issuers captures meaningful differences in market development and information availability that influence sovereign issuance decisions. This assumption could be violated if omitted common factors simultaneously drive both corporate and sovereign green bond timing, if reverse causality occurs where anticipated sovereign issuance influences corporate timing, or if selection effects mean that countries with stronger green preferences see both earlier corporate

[‖]Baseline survival and cumulative hazard functions for OECD and non-OECD strata, along with Schoenfeld residual tests confirming the proportional hazards assumption, are reported in Appendix B.3. Models estimated separately for OECD and non-OECD subsamples are reported in Appendix B.2 (Tables B.5, B.6, B.7, B.8) and yield substantively consistent results.

issuance and faster sovereign adoption. To address these concerns, several robustness checks are implemented.

6.2 Dependent Variable and Event Definition

The dependent variable in the survival analysis is the time until a country's first sovereign green bond issuance. Rather than analyzing the frequency or volume of sovereign green bond activity, the focus is on the timing of market entry, which represents a distinct policy decision with different underlying determinants than subsequent issuances. The event of interest is defined as a binary indicator equal to one in the year a country issues its first sovereign green bond and zero otherwise. Countries that do not issue a sovereign green bond during the observation period are treated as right-censored, meaning their survival time is observed through the end of the sample without experiencing the event. This approach captures the initial decision to enter the sovereign green bond market—the moment at which governments signal commitment to green financing and respond to informational and market-development signals generated by prior corporate green bond activity. Once a country issues its first sovereign green bond, it exits the risk set, as subsequent issuances reflect ongoing financing needs rather than the initial market entry decision.

6.3 Independent Variables

Green Bonds The primary variables of interest capture different dimensions of corporate green bond market development. Corporate private green bonds (*Corporate Private Cumulative*$_{t-1}$) measure cumulative issuance by private corporations, while corporate government green bonds (*Corporate Government Cumulative*$_{t-1}$) capture issuance by government-linked corporations. Financial sector bonds for both private and government-linked financial institutions (*Financial Private Cumulative*$_{t-1}$ and *Financial Government Cumulative*$_{t-1}$) are measured separately, capturing issuance by banks, credit unions, and other financial intermediaries whose primary business is providing financial services. The analysis also includes

issuance by other institutional issuers: agency bonds (*Agency Cumulative*$_{t-1}$), typically issued by national or regional government agencies or government-sponsored enterprises to finance public programs, and municipal bonds (*Municipal Cumulative*$_{t-1}$), issued by local or regional governments to finance infrastructure, utilities, and other community projects.

Distinguishing among these issuer types is important because their financing mandates, risk profiles, and political visibility differ, which may shape both their incentives to issue green bonds and the signaling effects such issuances have on sovereign and corporate markets. All corporate green bond variables are lagged by one period to address potential simultaneity concerns and allow for information processing and policy response time.

Other Control Variables *Total Climate Laws*$_{t-1}$ (Climate Policy Radar and Climate Change Laws of the World, 2025) is included as the total number of climate-related laws and policies in force in the previous year. This measure captures the breadth of a country's domestic climate policy framework, which may signal commitment to environmental action and influence the decision to issue green bonds. Also included is *Climate Disasters Total*$_{t-1}$ (Centre for Research on the Epidemiology of Disasters (CRED), 2025) as the total annual number of climate-related disasters—wildfires, droughts, extreme temperatures, floods, landslides, and storms—recorded in the previous year. This measure captures recent exposure to climate hazards, which may increase the salience of environmental financing and the likelihood of issuing green bonds. The analysis includes Treaty Cumulative Ratified$_{t-1}$ and Treaty Ratified$_{t-1}$, both constructed from the International Environmental Agreements Database Project (International Environmental Agreements Database Project, 2025) using only ratification actions after 1999. *Treaty Cumulative Ratified*$_{t-1}$ captures the total number of environmental agreements a country has ratified up to the prior year, while Treaty Ratified$_{t-1}$ is a binary indicator of whether the country ratified at least one agreement in the previous year.

The models include several controls capturing environmental endowments, financial de-

velopment, macroeconomic conditions, and domestic institutions. *Log(Forest Area*$_{t-1}$*)* (Food and Agriculture Organization of the United Nations, 2022) measures the extent of a country's forest resources; countries with larger forest areas may have stronger incentives to protect these resources, potentially increasing the likelihood of issuing green bonds. *Financial Development*$_{t-1}$, obtained from the International Monetary Fund's Financial Development Index database (International Monetary Fund, 2023), controls for differences in financial market development.

To account for variation in economic scale and performance, the models include GDP per capita (constant 2015 US$), GDP per capita growth (annual %), and total population—*Log(GDP per capita)*$_{t-1}$, *GDP Growth per capita*$_{t-1}$, and *Log(Population)*$_{t-1}$, respectively—sourced from the World Bank's World Development Indicators (World Bank, 2025). Domestic institutional factors are captured using variables from Coppedge et al. (2025). Across model specifications, these include *Electoral Power of Large Parties*$_{t-1}$, *Electoral Local Power*$_{t-1}$, and *Liberal Democracy*$_{t-1}$.

7 Empirical Results

7.1 Corporate Infrastructure and Sovereign Adoption

Tables 7 and 8 report results from Cox proportional hazards models stratified by OECD membership. Coefficients are expressed as hazard ratios, with values above one indicating increased likelihood of sovereign adoption and values below one indicating decreased likelihood. Because the cumulative issuance measures capture the long-term buildup of market infrastructure, each hazard ratio represents the effect of each additional bond added to the accumulated stock, with impacts compounding as issuance accumulates.

The results provide strong support for the infrastructure accumulation mechanism. *Corporate Private Cumulative*$_{t-1}$ is positively and significantly associated with sovereign adoption across specifications. Hazard ratios range from approximately 1.017 to 1.021, indicating

Table 7: Cox Proportional Hazards Models: Infrastructure Accumulation Effects

Variable	Model 1	Model 2	Model 3
Agency Cumulative$_{t-1}$	0.892**	0.936*	0.933*
	(0.039)	(0.033)	(0.034)
Municipal Cumulative$_{t-1}$	1.007	1.003	1.003
	(0.004)	(0.003)	(0.004)
Financial Private Cumulative$_{t-1}$	0.991	0.985	0.986
	(0.033)	(0.027)	(0.026)
Financial Government Cumulative$_{t-1}$	0.994	0.993	1.001
	(0.021)	(0.016)	(0.015)
Corporate Private Cumulative$_{t-1}$	1.021**	1.017**	1.017*
	(0.007)	(0.006)	(0.007)
Corporate Government Cumulative$_{t-1}$	1.017	1.009	1.006
	(0.013)	(0.01)	(0.01)
Climate Laws Total$_{t-1}$	1.003	0.978	0.971
	(0.036)	(0.036)	(0.037)
Climate Disasters Total$_{t-1}$	0.888	1.012	0.998
	(0.102)	(0.081)	(0.082)
Treaty Cumulative Ratified$_{t-1}$	0.948**	0.996	0.993
	(0.019)	(0.014)	(0.015)
Treaty Ratified$_{t-1}$	2.459**	2.085*	2.136*
	(0.306)	(0.321)	(0.312)
Log(Forest Area)$_{t-1}$	0.78*	0.878	0.916
	(0.111)	(0.099)	(0.092)
Log(GDP per cap.)$_{t-1}$	2.438*	2.605**	1.993*
	(0.352)	(0.309)	(0.293)
GDP Growth$_{t-1}$	0.974	0.979	0.982
	(0.024)	(0.021)	(0.022)
Log(Population)$_{t-1}$	2.394**	1.705*	1.621†
	(0.294)	(0.257)	(0.262)
Financial Development$_{t-1}$	0.111	0.114	0.194
	(1.664)	(1.431)	(1.435)
Electoral Power of Large Parties$_{t-1}$	1.218		
	(0.156)		
Electoral Local Power$_{t-1}$		1.014	
		(0.189)	
Liberal Democracy$_{t-1}$			2.962
			(1.01)
Observations	776	934	975
Events	37	46	48
Countries	79	96	99
Concordance	0.737	0.67	0.67

Note: Hazard ratios reported. Robust standard errors clustered by country in parentheses. All models stratified by OECD membership. *** p<0.001, ** p<0.01, * p<0.05, † p<0.10.

that each additional private corporate green bond in the prior period increases the instantaneous likelihood of sovereign issuance by about 1.7–2.1%.

While the per-bond effect appears modest, the cumulative nature of infrastructure accumulation generates substantial aggregate impacts. A country with ten additional private corporate bonds experiences an estimated 18–23% increase in the hazard of sovereign adoption, while fifty additional bonds correspond to roughly a doubling of the adoption hazard. These magnitudes align with the theoretical account: each corporate transaction incrementally expands the stock of available market infrastructure — including documentation templates, verification expertise, and investor familiarity — and sovereigns respond to the accumulated stock rather than individual issuances.

By contrast, *Agency Cumulative*$_{t-1}$ is associated with a statistically significant reduction in adoption likelihood, suggesting that not all forms of market development generate infrastructure that is equally transferable to sovereign issuers. Municipal and financial-sector issuance show no consistent relationship with sovereign adoption.

This pattern directly supports the bank-mediated diffusion mechanism documented in Section 5. Founding banks concentrated their early contracting activity on private corporations, using these transactions to develop the procedural templates and governance frameworks later codified in the GBP. The positive coefficient on private corporate issuance captures the downstream effect of this infrastructure on sovereign adoption decisions. As a robustness check, all models are re-estimated with time-varying coefficients that allow infrastructure effects to evolve as markets mature; results are substantively consistent and reported in Appendix B.2.**

**Augmenting the models with bank-level intermediation covariates and value-weighted issuance measures leaves the main findings unchanged. See Appendices D and E. Models excluding the ten political signaling countries identified in Section 2.2 also leave the main findings unchanged (Appendix F, Table F.1).

Table 8: Interpretation of Key Hazard Ratio Results

Variable	HR	Calculation	Effect
Agency Cumulative$_{t-1}$	0.892**	$(1 - 0.892) \times 100\% = 10.8\% \downarrow$	Reduces adoption likelihood
Corporate Private Cumulative$_{t-1}$	1.021**	$(1.021 - 1) \times 100\% = 2.1\% \uparrow$	Increases adoption likelihood

Note: HR = Hazard Ratio. ↑ indicates increased adoption likelihood; ↓ indicates decreased likelihood. Coefficients are from Model 1. ** p<0.01.

7.2 Agency Substitution Effects

In contrast to the infrastructure-building role of private corporate issuance, agency green bonds exhibit a significant negative association with sovereign adoption. Hazard ratios for *Agency Cumulative*$_{t-1}$ range from 0.947 to 0.97 across specifications, implying that each additional agency bond reduces the instantaneous likelihood of sovereign issuance by 3–5%. As with corporate effects, this impact compounds: ten agency bonds correspond to a 26–40% reduction in the hazard of sovereign adoption.

This substitution effect reflects a fundamentally different relationship between agency and sovereign issuance than between corporate and sovereign issuance. While corporate bonds build market infrastructure that sovereigns can access, agency bonds provide an alternative channel for governments to participate in green finance markets without direct sovereign issuance. Development banks, state-owned utilities, and government-sponsored enterprises can fulfill public financing objectives, satisfy domestic environmental constituencies, and signal climate commitment—all without requiring the political decision to issue sovereign green bonds.

Table 9 illustrates these dynamics through country cases. The United States represents the extreme scenario: with over 9,000 agency green bonds and more than 20,000 municipal bonds issued since 2012, the US has constructed a massive government-linked green bond infrastructure that appears to eliminate the need for sovereign issuance entirely.[††] Other countries show more moderate substitution. Japan accumulated 324 agency bonds before

[††]Given the potential for the U.S. to exert disproportionate influence on the agency substitution coefficient, all models are re-estimated excluding the United States; the agency substitution effect persists across all specifications (Appendix G, Table G.1).

issuing its first sovereign green bond in 2024—an 8-year delay from first agency activity. Australia waited 11 years between first agency issuance (2013) and sovereign adoption (2024).

Table 9: Agency Activity and Sovereign Adoption: Country Cases

Country	Agency Bonds Pre-Sovereign	First Agency Year	Sovereign Adopted?	Sovereign Year
United States	9379	2012	No	—
Japan	324	2016	Yes	2024
China	81	2016	Yes	2025
Norway	48	2013	No	—
Germany	33	2014	Yes	2020
Finland	16	2016	No	—
Netherlands	15	2014	Yes	2019
Australia	15	2013	Yes	2024
Sweden	14	2015	Yes	2020
Spain	12	2015	Yes	2021

Note: Agency bonds counted through year prior to sovereign entry; for non-adopters, through 2024.

Notably, some countries with substantial agency activity never adopt sovereign green bonds during the observation period. Norway (48 agency bonds since 2013) and Finland (16 agency bonds since 2016) appear to satisfy government green financing needs entirely through agency channels. These cases suggest that agency issuance can fully replace sovereign entry rather than merely postponing it.

The contrast between corporate and agency effects clarifies the mechanisms through which different issuer types influence sovereign decisions. Corporate issuance builds infrastructure that *facilitates* sovereign entry by lowering costs and uncertainty. Agency issuance *substitutes* for sovereign entry by providing alternative channels that satisfy the same underlying objectives. Both effects are statistically and substantively significant, but they operate through distinct causal pathways.

7.3 Reverse Causality

A natural concern is that the relationship between corporate and sovereign green bond issuance reflects common underlying preferences rather than infrastructure accumulation. Countries with stronger environmental commitments may attract both earlier corporate issuance and faster sovereign adoption, generating a spurious positive association. Alternatively, anticipated sovereign entry could stimulate corporate issuance if firms expect government programs to validate the market or expand the investor base—a pattern consistent with the conventional sovereign-first model of bond market development.

To assess these alternative explanations, the direction of the Cox models is reversed, using cumulative sovereign issuance to predict a country's first corporate green bond (full results in Appendix C, Table C.1). If sovereign activity catalyzes corporate adoption, sovereign issuance should significantly increase the corporate hazard. Instead, sovereign green bond issuance is associated with a *reduced* likelihood of subsequent corporate adoption, with hazard ratios ranging from 0.71 to 0.85 across specifications. Although none of these estimates reach conventional levels of statistical significance, the consistently negative direction is difficult to reconcile with either the common-preferences account—which predicts a positive association in both directions—or the sovereign-catalysis alternative. The pattern instead supports the interpretation that the primary causal direction runs from corporate infrastructure-building to sovereign entry, rather than the reverse.

These results do not eliminate all endogeneity concerns. Unobserved country-level characteristics could still jointly determine both corporate and sovereign timing through channels not captured by the controls. However, the failure of the reverse specification to produce the positive sovereign-to-corporate effect predicted by alternative explanations provides meaningful evidence against the most prominent identification threats.

8 Conclusion

This paper began with a puzzle: why do sovereign green bond markets emerge after private markets, and why does this pattern hold so consistently across countries with different levels of development, political systems, and environmental pressures? The evidence presented here points not to state incapacity or policy failure, but to the deliberate market-construction activities of global investment banks. Corporate green bond issuance matters not because governments imitate private actors, but because firms provide the organizational setting through which banks construct the institutional infrastructure—documentation standards, verification protocols, investor networks, and reputational benchmarks—that makes sovereign participation feasible. The four-year median lag between first corporate and first sovereign issuance is not an anomaly to be explained away; it reflects the time required to build market capacity compatible with sovereign political constraints.

The empirical findings are consistent across model specifications, sample restrictions, and estimation approaches. Corporate private green bond activity significantly accelerates sovereign adoption, with cumulative effects that are substantial: fifty additional corporate bonds nearly doubles the hazard of sovereign entry. Government agency issuance operates through an entirely different pathway, substituting for sovereign participation rather than facilitating it. The bank intermediation evidence complements these results: GBP founding banks underwrote 100% of private corporate green bonds before the principles existed, concentrated their early contracting activity on private rather than government-linked firms, and appear as lead managers in nearly three-quarters of countries' sovereign market entry deals. This pattern of concentrated early activity followed by declining market share as infrastructure became self-sustaining distinguishes the infrastructure accumulation mechanism from simple market power or competitive advantage.

The broader theoretical contribution is about how authority operates in global governance. The green bond case reveals a form of private authority that differs from the rule-making power documented in studies of credit rating agencies, accounting standards

bodies, or international financial institutions. Banks did not merely set rules that others followed; they built the market reality within which subsequent actors—including sovereign borrowers—had to operate. This constitutive dimension of financial intermediary power is undertheorized in international political economy, where private authority is more commonly analyzed as formal standard-setting or regulatory capture. The GBP consortium did not capture a regulator; it filled a governance vacuum, constructing the institutional infrastructure that no state or intergovernmental body was positioned to provide.

This has direct implications for debates about state autonomy in global environmental governance. Sovereign green bond programs are not simply financing instruments; they are political commitments that signal climate ambition to international audiences and domestic constituencies alike. Yet the terms on which governments can make these commitments—the standards they must meet, the verification providers they must engage, the investor expectations they must satisfy—were established through private corporate transactions that sovereigns had no role in shaping. States enter markets whose rules they did not write, under conditions constructed by private actors operating well before any government had decided to participate. This is not coercion; it is the structural power of market infrastructure, operating through the institutional conditions that define what legitimate participation looks like.

The substitution finding adds a further dimension to this picture. Countries whose public sector financing needs are met through government agencies—the United States being the extreme case—may never enter sovereign green bond markets at all, not because they lack capacity or commitment but because the institutional demand for direct sovereign participation has been satisfied through delegated entities. This suggests that the pathway through which market norms reach states is more contingent than diffusion theories typically acknowledge: the same infrastructure that facilitates entry for some governments may render entry unnecessary for others, depending on how public climate finance is organized domestically.

These findings carry practical implications for climate finance strategy. Efforts to acceler-

ate sovereign green bond adoption in countries where corporate markets remain nascent may be better directed at building the financial intermediary infrastructure—trained underwriters, established verification providers, familiar investor bases—than at directly incentivizing sovereign issuance. Conversely, in countries with large and well-developed agency green bond markets, the political and institutional case for direct sovereign participation may need to be made explicitly, since market infrastructure alone will not create the pressure for entry. The structure of financial intermediation, not just capital availability or political will, shapes whether and when sustainable finance norms reach sovereign borrowers.

More broadly, this paper identifies a form of private authority that IPE has not fully theorized. Existing accounts treat private authority primarily as rule-setting—credit rating agencies assigning creditworthiness, accounting standards bodies codifying disclosure—or as regulatory capture, where private actors colonize public institutions. The green bond case reveals a third form: market construction, in which financial intermediaries build the institutional reality within which states must subsequently operate, without capturing any regulator or formal standard-setting body. As new climate instruments emerge—biodiversity credits, transition bonds, debt-for-nature swaps—the question of who constructs the market infrastructure through which these instruments travel will shape state autonomy and governance outcomes in ways that neither the state-led nor market-pressure frameworks can anticipate. The green bond market offers an unusually well-documented instance of this dynamic, and the analytical tools developed here should travel.

References

Abbott, Kenneth W, Philipp Genschel, Duncan Snidal and Bernhard Zangl. 2021. Orchestration: Global governance through intermediaries. In *The Spectrum of International Institutions*. Routledge pp. 140–170.

Baker, Malcolm, Daniel Bergstresser, George Serafeim and Jeffrey Wurgler. 2021. "Financing the Response to Climate Change: The Pricing and Ownership of U.S. Green Bonds." *Journal of Financial Economics* 142(2):499–516.

Bank of America Merrill Lynch, Citi, Crédit Agricole CIB and JPMorgan Chase. 2014. "Green Bond Principles Created to Help Issuers and Investors Deploy Capital for Green Projects that Deliver Environmental Benefits." Press Release.
URL: *https://www.ca-cib.com/sites/default/files/2017-03/2014-01-13-cp-cacib-green-bond-principles-en-final.pdf*

Bénabou, Roland and Jean Tirole. 2010. "Individual and corporate social responsibility." *Economica* 77(305):1–19.

Box-Steffensmeier, Janet M. and Bradford S. Jones. 2004. *Event History Modeling: A Guide for Social Scientists*. Cambridge: Cambridge University Press.

Ceci, Marilyn. 2024. "Sustainability Warrior: From Creating Green Bond Principles to Financing the Energy Transition." SRI 360 Podcast. Interview transcript.
URL: *https://www.sri360.com*

Centre for Research on the Epidemiology of Disasters (CRED). 2025. "The Emergency Events Database (EM-DAT)." Online database. Accessed 2025-08-13.
URL: *https://www.emdat.be*

Claquin, Tanguy. 2015. "Green Bonds – A Promising Tool for Climate Finance." *Private Sector & Development* 22:6–8.
URL: *https://www.proparco.fr*

Climate Policy Radar and Climate Change Laws of the World. 2025. "Full Document Dataset." Online database. Data last updated 2025-06-16. Accessed 2025-08-13.
URL: *https://climatepolicyradar.notion.site/Readme-for-document-data-download-f2d55b7e238941b59559b9b1c4cc52c5*

Coppedge, Michael, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, David Altman, Fabio Angiolillo, Michael Bernhard, Agnes Cornell, M. Steven Fish, Linnea Fox, Lisa Gastaldi, Haakon Gjerløw, Adam Glynn, Ana Good God, Sandra Grahn, Allen Hicken, Katrin Kinzelbach, Joshua Krusell, Kyle L. Marquardt, Kelly McMann, Valeriya Mechkova, Juraj Medzihorsky, Natalia Natsika, Anja Neundorf, Pamela Paxton, Daniel Pemstein, Johannes von Römer, Brigitte Seim, Rachel Sigman, Svend-Erik Skaaning, Jeffrey Staton, Aksel Sundström, Marcus Tannenberg, Eitan Tzelgov, Yi-ting Wang, Felix Wiebrecht, Tore Wig, Steven Wilson and Daniel Ziblatt. 2025. "V-Dem Country-Year Dataset v15.".

Cutler, A. Claire, Virginia Haufler and Tony Porter, eds. 1999. *Private Authority and International Affairs*. Albany, NY: SUNY Press.

DiMaggio, Paul. 1988. Interest and Agency in Institutional Theory. In *Institutional Patterns and Organizations: Culture and Environment*, ed. Lynne G. Zucker. Cambridge, MA: Ballinger pp. 3–22.

Dittmar, Robert F and Kathy Yuan. 2008. "Do sovereign bonds benefit corporate bonds in emerging markets?" *The Review of Financial Studies* 21(5):1983–2014.

Drezner, Daniel W. 2008. "All Politics Is Global: Who Makes the Rules?" *International Organization* 62(3):389–410.

Durbin, Erik and David Ng. 2005. "The sovereign ceiling and emerging market corporate bond spreads." *Journal of international Money and Finance* 24(4):631–649.

Eckhart, Michael. 2021. "Climate Positive Podcast Interview." Climate Positive Podcast. Interview transcript.

Farrell, Henry and Abraham L Newman. 2019. "Weaponized interdependence: How global economic networks shape state coercion." *International security* 44(1):42–79.

Flammer, Caroline. 2021. "Corporate green bonds." *Journal of financial economics* 142(2):499–516.

Food and Agriculture Organization of the United Nations. 2022. "FAOSTAT Land, Inputs and Sustainability: Land Use." Online database. Accessed 2023-03-28; IMF staff calculations.
URL: *http://www.fao.org/faostat/en/data/RL*

Grundy, Bruce D, Sjoerd van Bekkum and Patrick Verwijmeren. 2024. "Complementarity of sovereign and corporate debt issuance: mind the gap." *Review of Finance* 28(4):1187–1213.

Hale, Thomas. 2020. "Catalytic cooperation." *Global Environmental Politics* 20(4):73–98.

Heimer, Matthew. 2018. "How Bank of America Financed a Green Energy Boom." *Fortune* .
URL: *https://fortune.com/2018/08/20/bank-of-america-wind-solar-energy/*

International Environmental Agreements Database Project. 2025. "International Environmental Agreements (IEA) Database Project." Online database, hosted by Université Laval; originally initiated at University of Oregon. Accessed 2025-08-11.
URL: *https://www.iea.ulaval.ca/en*

International Monetary Fund. 2023. "Financial Development Index Database." Online; legacy IM D Portal – accessed via legacydata.imf.org with key sk=f8032e80-b36c-43b1-ac26-493c5b1cd33b. Accessed 2025-08-13.
URL: *https://legacydata.imf.org/?sk=f8032e80-b36c-43b1-ac26-493c5b1cd33b*

Pistor, Katharina. 2019. *The Code of Capital: How the Law Creates Wealth and Inequality.* Princeton, NJ: Princeton University Press.

Presbitero, Andrea, Mr Dhaneshwar Ghura, Mr Olumuyiwa Adedeji and Lamin Njie. 2015. *International sovereign bonds by emerging markets and developing economies: drivers of issuance and spreads*. International Monetary Fund.

Riedl, Arno and Paul Smeets. 2017. "Why do investors hold socially responsible mutual funds?" *the Journal of Finance* 72(6):2505–2550.

Simmons, Beth A., Frank Dobbin and Geoffrey Garrett. 2006. "Introduction: The International Diffusion of Liberalism." *International Organization* 60(4):781–810.

Sinclair, Timothy J. 2005. *The New Masters of Capital: American Bond Rating Agencies and the Politics of Creditworthiness*. Ithaca, NY: Cornell University Press.

Tang, Dragon Yongjun and Yupu Zhang. 2020. "Do shareholders benefit from green bonds?" *Journal of Corporate Finance* 61:101427.

Tolliver, Clarence, Alexander Ryota Keeley and Shunsuke Managi. 2020. "Drivers of green bond market growth: The importance of Nationally Determined Contributions to the Paris Agreement and implications for sustainability." *Journal of cleaner production* 244:118643.

World Bank. 2025. "World Development Indicators." Online data retrieved from the World Development Indicators database via the World Bank DataBank. Accessed 2025-08-13. **URL:** *https://databank.worldbank.org/reports.aspx?source=2series=NY.GDP.MKTP.CD*

Zerbib, Olivier David. 2025. The effect of pro-environmental preferences on bond prices: Evidence from green bonds. In *Handbook of Quantitative Sustainable Finance*. Chapman and Hall/CRC pp. 410–448.

Appendix

When Firms Lead States: Financial Intermediaries and the Politics of Sovereign Green Bond Adoption

Table of Contents

A Descriptive Statistics **A1**

A.1 Bond Types A1

A.2 OECD vs. Non-OECD Corporate Adoption Rates A1

B Main Results **A4**

B.1 Cox Model Interpretation A4

B.2 Supplementary Models A5

B.3 Cox Tests A18

C Reverse Causality **A24**

D Robustness to Explicit Bank Intermediation Controls **A29**

E Robustness Check: Value-Weighted Cumulative Issuance **A31**

F Robustness Check: Excluding Political Signaling Countries **A33**

G Robustness Check: Excluding the United States **A35**

H Additional Evidence on Bank Motivations **A37**

H.1 The "Financed, Not Funded" Philosophy A37

H.2 Additional Statements from GBP Co-Authors A37

H.3 Reputational Risk Management A38

I Climate Treaty Amplification **A38**

A Descriptive Statistics

A.1 Bond Types

Table A.1: Green Bond Market by Bond Type, 2012–2024

Bond Type	N Issuances	Total Value (Bil. $)	First Year	Last Year
Pure Bond Types				
Green Bond	27,730	3,312.2	2012	2024
Social Bond	13,923	785.7	2013	2024
Sustainability Bond	5,985	590.8	2013	2024
Sustainability-Linked Bond	728	283.6	2015	2024
Transition Bond	97	35.3	2017	2024
Hybrid Bond Types				
Green + Social Bond	39	3.0	2021	2023
Green + Sustainability Bond	17	3.5	2018	2024
Social + Sustainability Bond	11	0.5	2018	2023
Green + Sustainability-Linked Bond	6	1.8	2021	2024
Sustainability + Sustainability-Linked Bond	3	1.0	2019	2024

Note:
Supranational issuers excluded. Total value reported in billions of USD.

A.2 OECD vs. Non-OECD Corporate Adoption Rates

Table A.2: Earliest Corporate Green Bond Issuance by OECD Status

OECD Status	Earliest Year
Non-OECD	2014
OECD	2012

Table A.3: Implementation Gap for Environmental Treaty Ratifiers

Classification	Value
Total Treaty Ratifiers	103
Ratified but No Bonds	52
Percent Ratified but No Bonds	50.5%

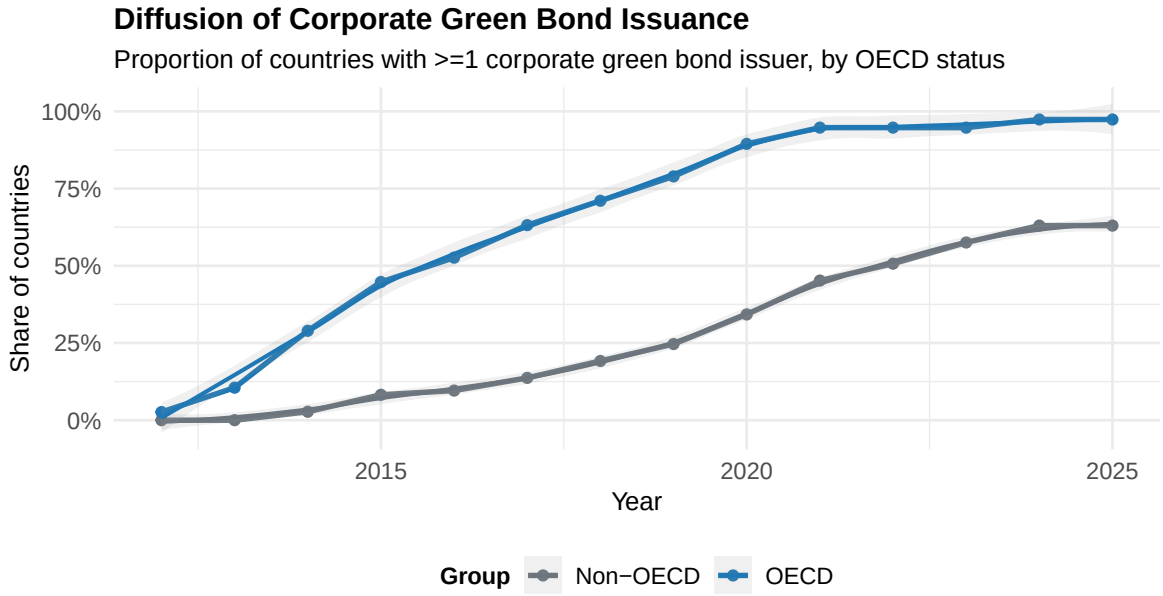
Notes: This table shows the implementation gap among countries in the green bond sample that ratified environmental treaties. Treaty ratifiers are defined as countries that have ratified at least one environmental treaty in the International Environmental Agreements Database during the sample period (2000-2024). "Ratified but No Bonds" indicates countries that ratified environmental treaties but never adopted sovereign green bonds. The 50.5% implementation failure rate demonstrates that international environmental commitments alone are insufficient for sovereign green bond adoption without underlying market foundations.

Table A.4: Paris Agreement Implementation Gap

Classification	Value
Total Paris Ratifiers	91
Ratified but No Bonds	45
Percent Ratified No Bonds	49.5%

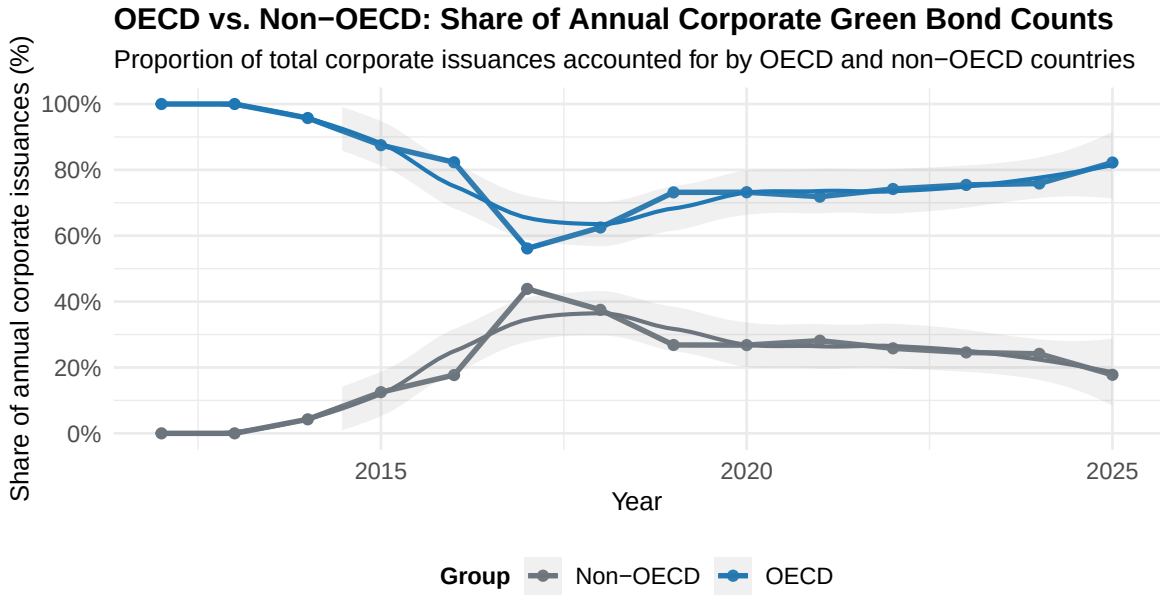
Notes: This table shows the implementation gap among countries in the green bond sample that ratified the Paris Agreement. "Ratified but No Bonds" indicates countries that ratified the Paris Agreement but never adopted sovereign green bonds. The 49.5% implementation failure rate demonstrates that international climate commitments alone are insufficient for sovereign green bond adoption.

Figure A.1



Note: Adoption = first year a country records a corporate green bond issuance.
The curve shows cumulative adoption, not issuance volume.

Figure A.2



Note: The figure shows the relative share of corporate green bond issuances (counts)
by OECD vs non−OECD countries each year. It reflects annual activity, not cumulative adoption.

B Main Results

B.1 Cox Model Interpretation

Table B.1: Hazard Ratio Interpretation Guide

Hazard Ratio	**Formula**	**Interpretation**
HR > 1	$(HR - 1) \times 100\%$	Percentage **increase** in hazard
HR < 1	$(1 - HR) \times 100\%$	Percentage **decrease** in hazard
HR $= 1$	No effect	No change in hazard

B.2 Supplementary Models

To assess whether the effects of accumulated market infrastructure evolve over time as countries gain experience observing corporate markets, the main Cox proportional hazards models are re-estimated with time-varying coefficients for key variables. Specifically, cumulative issuance variables, log GDP per capita, and financial development are interacted with the logarithm of time:

$$\mathrm{tt}(x, t) = x \times \log(t) \tag{2}$$

This specification allows the effect of cumulative issuance to change as market infrastructure matures, capturing potential learning effects or evolving market dynamics.

Tables B.2 and B.3 report results from these time-varying specifications. Overall, the findings remain substantively consistent with the baseline models.

Across specifications, *Corporate Private Cumulative*$_{t-1}$ remains positive and statistically significant, with hazard ratios ranging from approximately 1.007 to 1.01. These estimates indicate that each additional private corporate green bond increases the instantaneous likelihood of sovereign issuance by roughly 0.7–1.0%. Although smaller than the baseline estimates, this attenuation is expected: the time-varying specification distributes infrastructure effects across both the baseline and interaction terms.

Even modest per-bond effects generate meaningful aggregate impacts. Ten additional private corporate issuances correspond to roughly a 7–10% increase in adoption hazard, while fifty issuances imply increases of approximately 35–50%. These magnitudes remain consistent with the infrastructure accumulation mechanism, whereby repeated corporate transactions build documentation precedents, verification capacity, and investor familiarity that lower sovereign entry costs.

Agency issuance continues to exhibit a negative association with sovereign adoption. Hazard ratios for *Agency Cumulative*$_{t-1}$ range from approximately 0.95 to 0.97, indicating that each additional agency bond reduces the hazard of sovereign entry by roughly 3–5%. This pattern reinforces the substitution dynamic identified in the main text: agency financing can satisfy public climate investment needs without requiring direct sovereign issuance.

Because models using `tt()` incorporate time-dependent effects, survival curves cannot be generated using `survfit()`, which assumes proportional hazards. Inference therefore relies on estimated hazard ratios and diagnostic tests reported in Section B.3, which confirm that the time-interaction structure appropriately relaxes the proportional hazards assumption.

Table B.2: Cox Proportional Hazards Models: Infrastructure Accumulation Effects

Variable	Model 1	Model 2	Model 3
Agency Cumulative$_{t-1}$	0.951**	0.971*	0.969*
	(0.016)	(0.014)	(0.015)
Municipal Cumulative$_{t-1}$	1.004	1.002	1.002
	(0.002)	(0.002)	(0.002)
Financial Private Cumulative$_{t-1}$	0.994	0.992	0.993
	(0.013)	(0.011)	(0.011)
Financial Government Cumulative$_{t-1}$	0.997	0.997	1.001
	(0.01)	(0.007)	(0.007)
Corporate Private Cumulative$_{t-1}$	1.009**	1.008**	1.008**
	(0.003)	(0.003)	(0.003)
Corporate Government Cumulative$_{t-1}$	1.008	1.004	1.003
	(0.006)	(0.005)	(0.005)
Climate Laws Total$_{t-1}$	1.001	0.974	0.967
	(0.037)	(0.037)	(0.038)
Climate Disasters Total$_{t-1}$	0.881	1.003	0.989
	(0.101)	(0.081)	(0.082)
Treaty Cumulative Ratified$_{t-1}$	0.949**	0.996	0.993
	(0.019)	(0.014)	(0.015)
Treaty Ratified$_{t-1}$	2.523**	2.127*	2.172*
	(0.305)	(0.322)	(0.314)
Log(Forest Area)$_{t-1}$	0.776*	0.877	0.92
	(0.11)	(0.098)	(0.093)
Log(GDP per cap.)$_{t-1}$	1.505**	1.544**	1.367*
	(0.158)	(0.141)	(0.131)
GDP Growth$_{t-1}$	0.974	0.978	0.982
	(0.025)	(0.021)	(0.022)
Log(Population)$_{t-1}$	2.447**	1.717*	1.624†
	(0.299)	(0.259)	(0.264)
Financial Development$_{t-1}$	0.395	0.402	0.508
	(0.736)	(0.64)	(0.644)
Electoral Power of Large Parties$_{t-1}$	1.213		
	(0.154)		
Electoral Local Power$_{t-1}$		1.008	
		(0.188)	
Liberal Democracy$_{t-1}$			2.821
			(1.005)
Observations	776	934	975
Events	37	46	48
Countries	79	96	99
Concordance	0.738	0.673	0.673

Note: Hazard ratios reported. Robust standard errors clustered by country in parentheses. All models stratified by OECD membership. Time-varying coefficients applied to cumulative issuance variables, log GDP per capita, and financial development. *** p<0.001, ** p<0.01, * p<0.05, † p<0.10.

Table B.3 presents reduced-form models with time-varying effects applied to cumulative issuance variables and key macroeconomic controls. Results mirror the main specification.

Corporate Private Cumulative$_{t-1}$ remains positively associated with sovereign adoption (HR $\approx$ 1.008–1.009, $p < 0.01$ to $p < 0.001$). Each additional issuance increases the hazard by roughly 0.8–0.9%, implying that five additional bonds raise adoption likelihood by approximately 4–5% and ten bonds by 8–9%.

Agency Cumulative$_{t-1}$ remains negative and statistically significant (HR $\approx$ 0.955–0.97), indicating a 3–4.5% reduction in adoption hazard per issuance. This supports the substitution mechanism whereby agency-level financing delays sovereign market entry.

Treaty Ratified$_{t-1}$ continues to exhibit a strong positive association (HR $\approx$ 2.1–2.6), indicating that ratifying a relevant climate treaty more than doubles the likelihood of sovereign issuance. This underscores the catalytic role of international commitments in accelerating climate finance adoption.

Table B.3: Cox Model Comparison - Reduced w TV

Variable	Model 1	Model 2	Model 3
Agency Cumulative$_{t-1}$	0.955**	0.965*	0.97**
	(0.015)	(0.014)	(0.011)
Municipal Cumulative$_{t-1}$	1.004	1.002	1.002
	(0.003)	(0.002)	(0.002)
Financial Private Cumulative$_{t-1}$	0.992	0.998	0.995
	(0.014)	(0.009)	(0.008)
Financial Government Cumulative$_{t-1}$	0.998	1.005	1.002
	(0.01)	(0.007)	(0.006)
Corporate Private Cumulative$_{t-1}$	1.008**	1.009***	1.008***
	(0.003)	(0.003)	(0.002)
Corporate Government Cumulative$_{t-1}$	1.007	1.001	1.002
	(0.006)	(0.005)	(0.004)
Climate Laws Total$_{t-1}$	0.998	0.981	
	(0.036)	(0.038)	
Climate Disasters Total$_{t-1}$	0.869	1.016	
	(0.101)	(0.072)	
Treaty Cumulative Ratified$_{t-1}$	0.949**	0.994	
	(0.019)	(0.015)	
Treaty Ratified$_{t-1}$	2.566**	2.114*	
	(0.302)	(0.292)	
Log(Forest Area)$_{t-1}$	0.776*	0.951	
	(0.106)	(0.076)	
Log(GDP per cap.)$_{t-1}$	1.314**		
	(0.104)		
GDP Growth$_{t-1}$	0.979		
	(0.024)		
Log(Population)$_{t-1}$	2.182**		
	(0.252)		
Electoral Power of Large Parties$_{t-1}$	1.206		
	(0.15)		
Observations	776	994	1095
Events	37	50	55
Countries	79	101	111
Concordance	0.733	0.674	0.562

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10. Robust standard errors in parentheses below Hazard Ratios.

Table B.4 presents results from reduced-form Cox proportional hazards models without time-varying effects. The cumulative issuance measures capture the long-term buildup of market activity rather than a one-time change. Hazard ratios therefore represent the effect of *each additional issuance to date*, with the impact compounding as issuances accumulate. This means that even small per-unit effects can translate into substantial changes in the likelihood of sovereign issuance for countries with more active markets.

Corporate Private Cumulative$_{t-1}$ is positively and consistently associated with sovereign adoption across all specifications, with hazard ratios between 1.017 and 1.020 ($p < 0.01$ to $p < 0.001$). These values indicate that each additional private corporate issuance in the prior period increases the instantaneous likelihood of sovereign issuance by roughly 1.7–2.0%. While the per-unit effect is modest, the cumulative nature means that five additional issuances correspond to an estimated 8.5–10% higher hazard, and ten issuances to a 17–20% higher hazard. This pattern is consistent with a infrastructure accumulation, where sustained private-sector activity signals feasibility and viability for sovereign issuers.

Agency Cumulative$_{t-1}$ issuance shows a negative and statistically significant association with sovereign issuance, with hazard ratios ranging from 0.90 ($p < 0.01$) to 0.934 ($p < 0.05$). These results imply that each additional agency issuance in the prior period reduces the hazard of sovereign issuance by about 6.6–10%, with the effect compounding as agency issuance accumulates. This finding supports a substitution effect, in which sustained agency-level financing delays or displaces sovereign entry into the market.

Treaty Ratified$_{t-1}$ remains one of the strongest predictors, with hazard ratios between 2.09 ($p < 0.01$) and 2.49 ($p < 0.01$), indicating that ratifying a relevant climate treaty more than doubles the instantaneous likelihood of a first sovereign issuance. This underscores the role of international political commitments in accelerating domestic adoption of climate finance instruments.

Table B.4: Cox Model Comparison - Reduced

Variable	Model 1	Model 2	Model 3
Agency Cumulative$_{t-1}$	0.9**	0.925*	0.934**
	(0.037)	(0.033)	(0.025)
Municipal Cumulative$_{t-1}$	1.006	1.004	1.003
	(0.005)	(0.004)	(0.004)
Financial Private Cumulative$_{t-1}$	0.986	0.997	0.991
	(0.034)	(0.023)	(0.02)
Financial Government Cumulative$_{t-1}$	0.997	1.011	1.006
	(0.021)	(0.015)	(0.011)
Corporate Private Cumulative$_{t-1}$	1.018**	1.02**	1.017***
	(0.006)	(0.006)	(0.004)
Corporate Government Cumulative$_{t-1}$	1.014	1.001	1.003
	(0.013)	(0.011)	(0.008)
Climate Laws Total$_{t-1}$	1.001	0.984	
	(0.035)	(0.037)	
Climate Disasters Total$_{t-1}$	0.876	1.023	
	(0.101)	(0.07)	
Treaty Cumulative Ratified$_{t-1}$	0.949**	0.994	
	(0.019)	(0.015)	
Treaty Ratified$_{t-1}$	2.486**	2.093*	
	(0.303)	(0.292)	
Log(Forest Area)$_{t-1}$	0.78*	0.952	
	(0.107)	(0.076)	
Log(GDP per cap.)$_{t-1}$	1.75*		
	(0.233)		
GDP Growth$_{t-1}$	0.98		
	(0.024)		
Log(Population)$_{t-1}$	2.118**		
	(0.246)		
Electoral Power of Large Parties$_{t-1}$	1.208		
	(0.151)		
Observations	776	994	1095
Events	37	50	55
Countries	79	101	111
Concordance	0.729	0.671	0.559

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10. Robust standard errors in parentheses below Hazard Ratios.

Table B.5 presents results from Cox proportional hazards models with time-varying effects for the subsample of non-OECD countries. The cumulative issuance variables capture the

long-term buildup of market activity rather than a one-time change, meaning that hazard ratios reflect the effect of *each additional issuance to date*, with the impact compounding as issuances accumulate. Even modest per-unit effects can therefore translate into substantial differences in the likelihood of sovereign issuance for countries with more active markets.

Table B.5: Cox Model Comparison - Non OECD w TV

Variable	Model 1	Model 2
Agency Cumulative$_{t-1}$	0.885	0.888
	(0.145)	(0.127)
Municipal Cumulative$_{t-1}$	1.11*	1.04
	(0.046)	(0.073)
Financial Private Cumulative$_{t-1}$	0.978	0.979
	(0.019)	(0.018)
Financial Government Cumulative$_{t-1}$	1.009	1.001
	(0.016)	(0.021)
Corporate Private Cumulative$_{t-1}$	1.01†	1.007**
	(0.006)	(0.002)
Corporate Government Cumulative$_{t-1}$	1.008	1.019
	(0.021)	(0.014)
Climate Laws Total$_{t-1}$	0.924	
	(0.087)	
Climate Disasters Total$_{t-1}$	1.027	
	(0.086)	
Treaty Cumulative Ratified$_{t-1}$	0.961	
	(0.031)	
Treaty Ratified$_{t-1}$	1.786	
	(0.425)	
Observations	677	749
Events	26	29
Countries	75	77
Concordance	0.74	0.588

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10. Robust standard errors in parentheses below Hazard Ratios.

In this subsample, Corporate Private Cumulative$_{t-1}$ is positively associated with sovereign adoption, with hazard ratios of 1.01 ($p < 0.10$) in Model 1 and 1.007 ($p < 0.01$) in Model 2. These results imply that each additional private corporate issuance in the prior period increases the hazard of sovereign issuance by approximately 0.7–1%. While modest in size, the cumulative nature means that five additional issuances correspond to an estimated 3.5–5% higher hazard, and ten issuances to a 7–10% higher hazard, consistent with a infrastructure accumulation from sustained private-sector activity.

By contrast, Agency Cumulative$_{t-1}$ is not statistically significant in either model, with hazard ratios around 0.885–0.888, suggesting no robust evidence of a substitution effect from

agency issuance in the non-OECD sample.

Treaty Ratified$_{t-1}$ is also not statistically significant in this restricted sample, with a hazard ratio of 1.786, although the direction of the association remains positive. This may indicate that the catalytic effect of treaty ratification on sovereign issuance is weaker or less precisely estimated among non-OECD issuers, potentially due to smaller sample size and lower overall issuance activity.

Table B.6 presents results from Cox proportional hazards models without time-varying effects for the subsample of non-OECD countries. The cumulative issuance variables capture the long-term buildup of market activity rather than a one-time change, meaning that hazard ratios reflect the effect of *each additional issuance to date*, with the impact compounding as issuances accumulate. Even modest per-unit effects can therefore translate into substantial differences in the likelihood of sovereign issuance for countries with more active markets.

Table B.6: Cox Model Comparison - Non OECD

Variable	Model 1	Model 2
Agency Cumulative$_{t-1}$	0.743	0.765
	(0.335)	(0.253)
Municipal Cumulative$_{t-1}$	1.264*	1.092
	(0.104)	(0.161)
Financial Private Cumulative$_{t-1}$	0.949	0.953
	(0.042)	(0.04)
Financial Government Cumulative$_{t-1}$	1.021	1.005
	(0.034)	(0.037)
Corporate Private Cumulative$_{t-1}$	1.021	1.015**
	(0.013)	(0.006)
Corporate Government Cumulative$_{t-1}$	1.018	1.042
	(0.048)	(0.031)
Climate Laws Total$_{t-1}$	0.931	
	(0.083)	
Climate Disasters Total$_{t-1}$	1.034	
	(0.087)	
Treaty Cumulative Ratified$_{t-1}$	0.961	
	(0.031)	
Treaty Ratified$_{t-1}$	1.773	
	(0.422)	
Observations	677	749
Events	26	29
Countries	75	77
Concordance	0.747	0.588

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10. Robust standard errors in parentheses below Hazard Ratios.

Corporate Private Cumulative$_{t-1}$ is positively associated with sovereign adoption in both models, with hazard ratios of 1.021 in Model 1 ($p < 0.10$) and 1.015 in Model 2 ($p < 0.01$). These results indicate that each additional private corporate issuance in the prior period increases the hazard of sovereign issuance by approximately 1.5–2.1%. While the per-unit effect is modest, the cumulative nature means that five additional issuances correspond to an estimated 7.5–10.5% higher hazard, and ten issuances to a 15–21% higher hazard, consistent with a infrastructure accumulation from sustained private-sector activity.

Agency Cumulative$_{t-1}$ is not statistically significant in either model, with hazard ratios of 0.743 and 0.765, suggesting no robust evidence of a substitution effect from agency issuance among non-OECD countries.

Treaty Ratified$_{t-1}$ is also not statistically significant in this subsample, with hazard ratios of 1.773 in both models, though the positive direction suggests that the effect of treaty ratification may still be meaningful but less precisely estimated for non-OECD issuers, likely due to smaller sample size and lower issuance activity.

Table B.7 presents results from Cox proportional hazards models with time-varying effects for the subsample of OECD countries. The cumulative issuance variables capture the long-term buildup of market activity rather than a one-time change, meaning that hazard ratios reflect the effect of *each additional issuance to date*, with the impact compounding as issuances accumulate. Even modest per-unit effects can therefore translate into substantial differences in the likelihood of sovereign issuance for countries with more active markets.

Table B.7: Cox Model Comparison - OECD w TV

Variable	Model 1	Model 2
Agency Cumulative$_{t-1}$	0.94*	0.934**
	(0.03)	(0.022)
Municipal Cumulative$_{t-1}$	1.002	1.004**
	(0.003)	(0.001)
Financial Private Cumulative$_{t-1}$	0.979†	0.985
	(0.011)	(0.01)
Financial Government Cumulative$_{t-1}$	1.139***	1.133***
	(0.037)	(0.032)
Corporate Private Cumulative$_{t-1}$	1.013†	1.014*
	(0.007)	(0.006)
Corporate Government Cumulative$_{t-1}$	0.996	0.994
	(0.006)	(0.005)
Climate Laws Total$_{t-1}$	0.998	
	(0.053)	
Climate Disasters Total$_{t-1}$	1.103	
	(0.119)	
Treaty Cumulative Ratified$_{t-1}$	1.009	
	(0.023)	
Treaty Ratified$_{t-1}$	2.129	
	(0.464)	
Observations	346	346
Events	26	26
Countries	36	36
Concordance	0.649	0.597

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10. Robust standard errors in parentheses below Hazard Ratios.

Corporate Private Cumulative$_{t-1}$ is positively associated with sovereign adoption, with hazard ratios of 1.013 ($p < 0.10$) in Model 1 and 1.014 ($p < 0.05$) in Model 2. These estimates imply that each additional private corporate issuance in the prior period increases the hazard of sovereign issuance by roughly 1.3–1.4%. While modest in size, the cumulative nature means that five additional issuances correspond to an estimated 6.5–7% higher hazard, and ten issuances to a 13–14% higher hazard, consistent with a infrastructure accumulation from sustained private-sector activity.

Agency Cumulative$_{t-1}$ is negatively associated with sovereign issuance in both models, with hazard ratios of 0.94 ($p < 0.05$) and 0.934 ($p < 0.01$), indicating that each additional agency issuance reduces the hazard by approximately 6–6.6%, with the effect compounding as agency issuance accumulates. This suggests a substitution effect in which agency-level financing may displace or delay sovereign issuance.

Treaty Ratified$_{t-1}$ is positively associated with sovereign issuance (HR = 2.129), indicating that treaty ratification more than doubles the instantaneous likelihood of a first sovereign issuance, although the result is not statistically significant in this subsample, potentially due to the smaller number of OECD countries and events.

A notable finding in the OECD sample is that Financial Government Cumulative$_{t-1}$ is strongly and positively associated with sovereign issuance (HR $\approx$ 1.133–1.139, $p < 0.001$), suggesting that sustained issuance by government-linked financial institutions may play an important role in facilitating sovereign entry into the green bond market.

Table B.8 presents results from Cox proportional hazards models without time-varying effects for the subsample of OECD countries. The cumulative issuance variables capture the long-term buildup of market activity rather than a one-time change, meaning that hazard ratios reflect the effect of *each additional issuance to date*, with the impact compounding as issuances accumulate. Even modest per-unit effects can therefore translate into substantial differences in the likelihood of sovereign issuance for countries with more active markets.

Table B.8: Cox Model Comparison - OECD

Variable	Model 1	Model 2
Agency Cumulative$_{t-1}$	0.878*	0.863**
	(0.062)	(0.045)
Municipal Cumulative$_{t-1}$	1.004	1.007**
	(0.004)	(0.002)
Financial Private Cumulative$_{t-1}$	0.958	0.97
	(0.029)	(0.025)
Financial Government Cumulative$_{t-1}$	1.308***	1.299***
	(0.079)	(0.07)
Corporate Private Cumulative$_{t-1}$	1.028†	1.03*
	(0.016)	(0.012)
Corporate Government Cumulative$_{t-1}$	0.991	0.986
	(0.013)	(0.011)
Climate Laws Total$_{t-1}$	0.998	
	(0.052)	
Climate Disasters Total$_{t-1}$	1.11	
	(0.119)	
Treaty Cumulative Ratified$_{t-1}$	1.01	
	(0.023)	
Treaty Ratified$_{t-1}$	2.069	
	(0.456)	
Observations	346	346
Events	26	26
Countries	36	36
Concordance	0.647	0.599

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10. Robust standard errors in parentheses below Hazard Ratios.

Corporate Private Cumulative$_{t-1}$ is positively associated with sovereign adoption, with hazard ratios of 1.028 ($p < 0.10$) in Model 1 and 1.03 ($p < 0.01$) in Model 2. These values indicate that each additional private corporate issuance in the prior period increases the hazard of sovereign issuance by roughly 2.8–3.0%. Given the cumulative nature of the variable, five additional issuances correspond to an estimated 14–15% higher hazard, and ten issuances to a 28–30% higher hazard, consistent with a strong infrastructure accumulation from sustained private-sector activity.

Agency Cumulative$_{t-1}$ is negatively associated with sovereign issuance in both models, with hazard ratios of 0.878 ($p < 0.05$) and 0.863 ($p < 0.01$). These results imply that each additional agency issuance reduces the hazard of sovereign issuance by about 12.2–13.7%, with the effect compounding as agency issuance accumulates. This suggests a substitution effect in which agency-level financing may displace or delay sovereign issuance.

Treaty Ratified$_{t-1}$ is positively associated with sovereign issuance (HR = 2.069), indicating that treaty ratification more than doubles the instantaneous likelihood of a first sovereign issuance. However, the result is not statistically significant in this subsample, likely due to the smaller sample size and limited number of events among OECD countries.

A notable and consistent finding in the OECD sample is that Financial Government Cumulative$_{t-1}$ is strongly and positively associated with sovereign issuance (HR $\approx$ 1.299–1.308, $p < 0.001$), suggesting that sustained issuance by government-linked financial institutions may play a particularly important role in facilitating sovereign entry into the green bond market.

B.3 Cox Tests

B.3.1 Test Results - Table 7 Model 1

Figure B.1 plots model-based baseline survival functions for OECD and non-OECD countries. The survival curves represent the probability that a country has not yet issued a sovereign green bond by each time point, conditional on the covariates included in the Cox model. Both curves remain close to one early in the period, reflecting that most countries are still "at risk" of first issuance. Over time, the OECD curve declines more rapidly, indicating that OECD members tend to adopt sovereign green bonds earlier than their non-OECD counterparts. The dashed lines are 95% confidence intervals, which widen as the number of countries at risk declines. The table beneath the figure reports the number of country-years at risk as well as the number of unique countries at risk in each stratum.

Figure B.2 shows the corresponding cumulative hazard functions, which emphasize differences in the risk of first issuance over time. The cumulative hazard rises more steeply for OECD members, suggesting that the instantaneous rate of adoption is higher relative to non-OECD states. This representation makes the divergence between groups more visible than in the survival curves, where probabilities remain near one. As with the survival plot, the risk table indicates the number of observations contributing to the hazard estimates at each time point.

Figure B.1

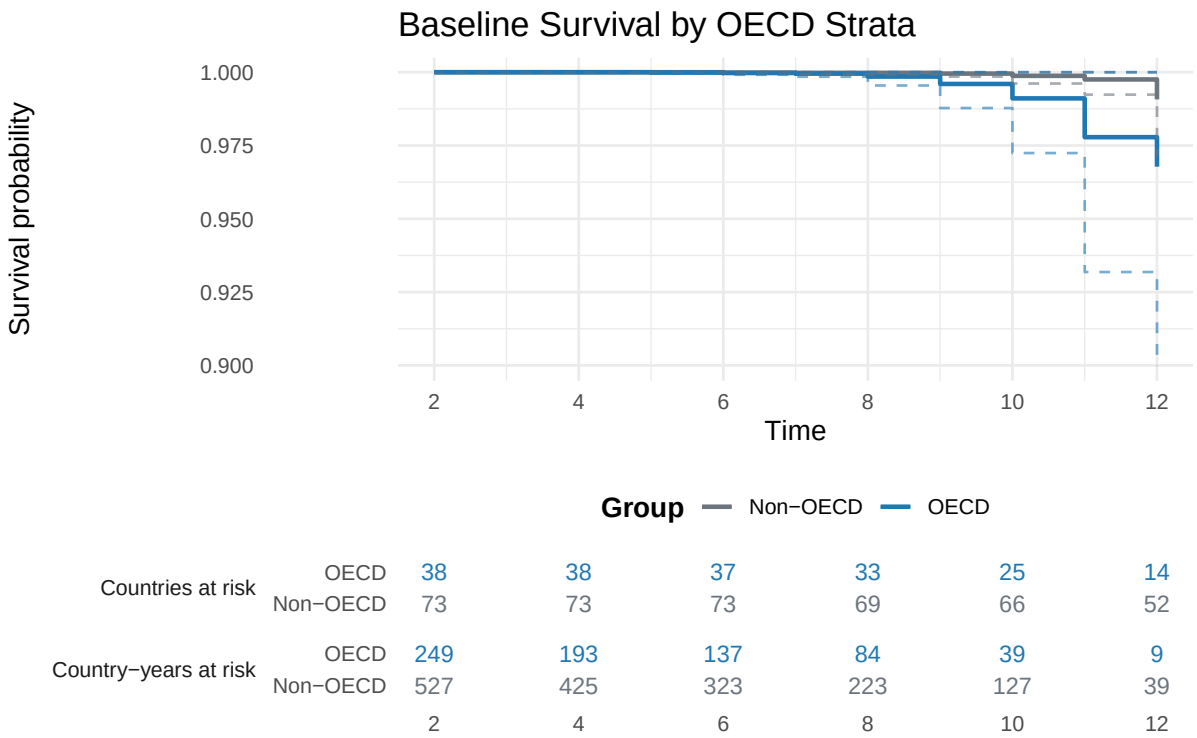


Figure B.2

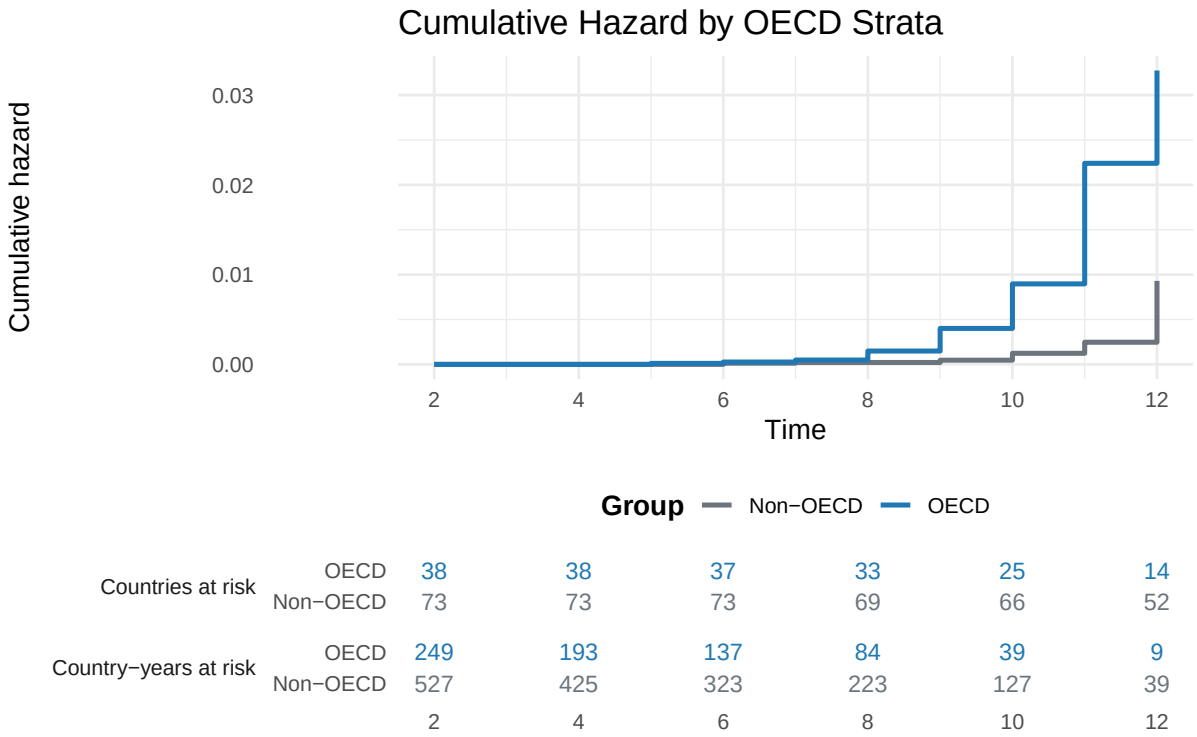


Table B.9: Tests of the Proportional Hazards Assumption (Schoenfeld Residuals)

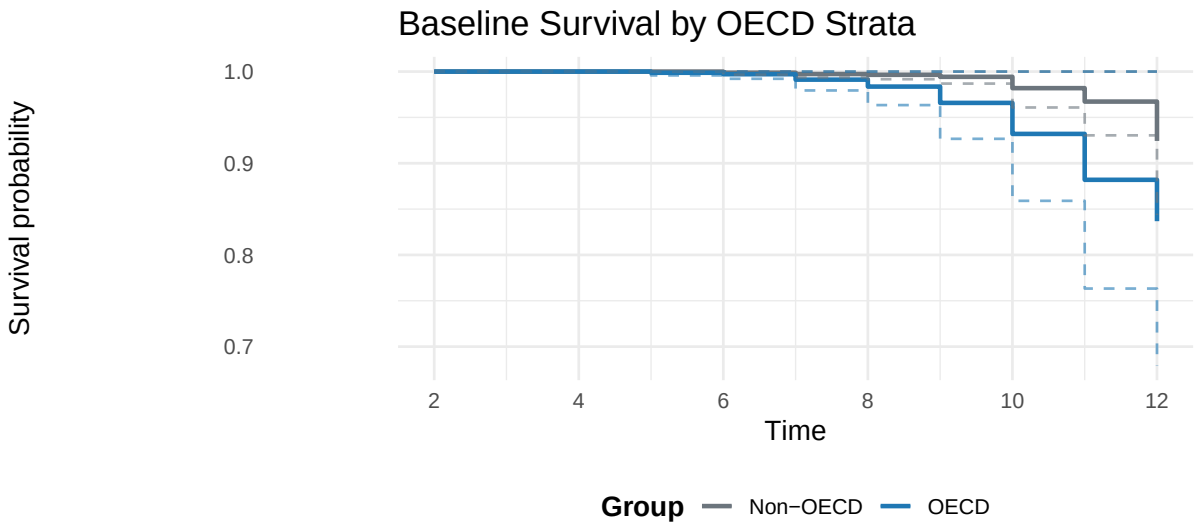
Variable	χ^2	df	p-value
Agency Cumulative$_{t-1}$	0.663	1	0.416
Municipal Cumulative$_{t-1}$	0.000	1	0.996
Financial Private Cumulative$_{t-1}$	0.209	1	0.648
Financial Government Cumulative$_{t-1}$	0.040	1	0.841
Corporate Private Cumulative$_{t-1}$	0.833	1	0.361
Corporate Government Cumulative$_{t-1}$	0.161	1	0.688
Electoral Power of Large Parties$_{t-1}$	0.141	1	0.707
Climate Laws Total$_{t-1}$	0.734	1	0.392
Climate Disasters Total$_{t-1}$	0.507	1	0.476
Treaty Cumulative Ratified$_{t-1}$	0.062	1	0.804
Treaty Ratified$_{t-1}$	3.51	1	0.061
Log(Forest Area)$_{t-1}$	0.667	1	0.414
Log(GDP per cap.)$_{t-1}$	**7.81**	1	**0.005**
GDP Growth$_{t-1}$	0.113	1	0.737
Log(Population)$_{t-1}$	0.071	1	0.790
Financial Development$_{t-1}$	2.00	1	0.158
GLOBAL test	16.4	16	0.422

The proportional hazards assumption holds for nearly all covariates in the model. The global test statistic is insignificant ($p = 0.422$), indicating no systematic violation across the model. At the variable level, GDP per capita (log) shows some evidence of non-proportionality ($p = 0.005$). All other covariates have non-significant tests, suggesting their effects on the hazard of first issuance are constant over time. These results provide reassurance that the proportional hazards assumption is broadly satisfied, with only limited evidence of deviation for GDP per capita.

B.3.2 Test Results - Table B4 Model 3

The baseline survival curves in Figure B.3 display the probability that a country has not yet issued a sovereign green bond over time, conditional on covariates in Model 2. Both OECD and non-OECD countries start near a survival probability of 1, reflecting that most remain "at risk" of issuance in the early years. Over time, the OECD curve declines more steeply than the non-OECD curve, indicating earlier and more frequent issuance among OECD members. Confidence intervals (not shown here) suggest that differences, while modest in magnitude, are consistent. The risk tables below the plot report both country-years at risk (rows in the panel) and unique countries at risk, making clear that OECD countries tend to leave the risk set sooner than non-OECD states.

Figure B.3

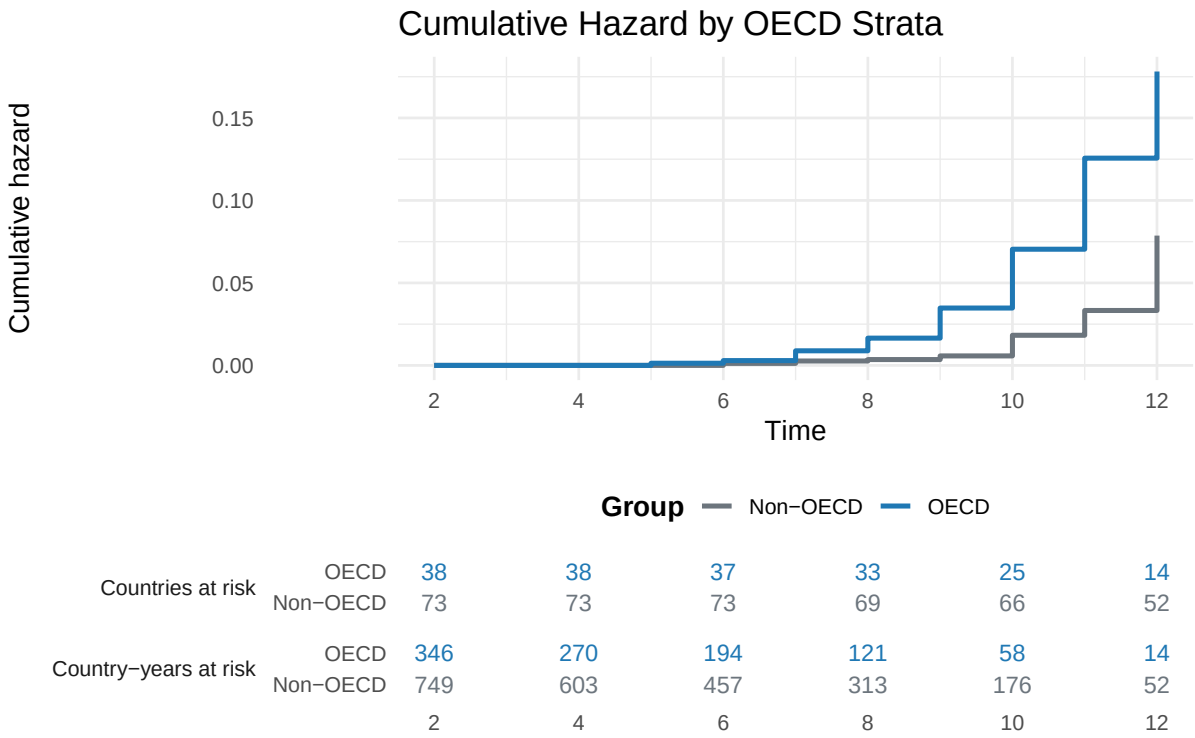


Countries at risk	OECD	38	38	37	33	25	14
	Non−OECD	73	73	73	69	66	52
Country−years at risk	OECD	346	270	194	121	58	14
	Non−OECD	749	603	457	313	176	52
		2	4	6	8	10	12

The cumulative hazard functions in Figure B.4 emphasize the timing of issuance risk. OECD members accumulate hazard more rapidly, showing a steeper increase over time relative to non-OECD countries. This implies that the instantaneous risk of first issuance is consistently higher among OECD states. The cumulative hazard representation makes group differences clearer than in the survival probability plot, where probabilities remain close to one. The risk tables again confirm that, as time progresses, relatively fewer OECD countries remain at risk compared to non-OECD peers.

Table B.10 reports the results from tests of the proportional hazards assumption based on Schoenfeld residuals. For each covariate, the null hypothesis is that the effect of the covariate on the hazard rate is proportional over time. All individual covariates have large p-values (0.60–0.91), indicating no evidence of time-varying effects. The global test similarly

Figure B.4



yields a non-significant result ($\chi^2 = 3.784$, p=0.71), confirming that the proportional hazards assumption holds for the model overall. This suggests that the estimated hazard ratios can be interpreted as constant multiplicative effects on the hazard of first sovereign green bond issuance across the observation period.

Table B.10: Proportional Hazards Assumption Test (Schoenfeld Residuals)

Variable	χ^2	df	p-value
Agency Cumulative$_{t-1}$	0.271	1	0.600
Municipal Cumulative$_{t-1}$	0.013	1	0.910
Financial Private Cumulative$_{t-1}$	0.034	1	0.850
Financial Government Cumulative$_{t-1}$	0.017	1	0.900
Corporate Private Cumulative$_{t-1}$	0.253	1	0.610
Corporate Government Cumulative$_{t-1}$	0.041	1	0.840
GLOBAL test	3.784	6	0.710

B.3.3 Time-Varying Models

For the specification including time–interaction terms (tt()), survival and cumulative hazard plots could not be generated because survfit() does not support Cox models with time–dependent effects. In this case, inference relies on the estimated hazard ratios reported in the tables and on the diagnostic tests already conducted, which confirm that the proportional hazards assumption is appropriately relaxed through the time–interaction structure. These results indicate that the extended model provides a valid representation of the underlying process, even though visualization through Kaplan–Meier or Cox survival curves is not available.

C Reverse Causality

Table C.1 reports Cox proportional hazards models testing a reverse-causality account: whether sovereign green bond issuance is followed by increased corporate issuance. Model 1 estimates the hazard of corporate green bond issuance in the full specification, Model 2 presents a closely related specification with minor adjustments to the covariate set and sample, and Model 3 reports a more parsimonious model.

Across all specifications, *Sovereign Cumulative*$_{t-1}$ is not positively associated with corporate adoption. The estimated hazard ratios are below one in each model (0.792, 0.854, and 0.711), indicating that additional sovereign issuance is associated with a lower subsequent hazard of corporate issuance. Substantively, these point estimates correspond to reductions in the instantaneous likelihood of corporate issuance of approximately 14.6–28.9%, although the estimates are not statistically significant. Taken together, the results do not support the claim that sovereign adoption systematically catalyzes corporate green bond issuance.

Other market segments exhibit heterogeneous associations with corporate adoption. *Agency Cumulative*$_{t-1}$ is consistently above one (HR $\approx$ 1.255–1.434, marginally significant in Models 2 and 3), suggesting that agency issuance may be complementary with corporate market activity. Municipal issuance is not associated with corporate adoption in Models 1–2 (HR $<$ 1), but becomes strongly positive in the parsimonious specification (HR = 1.794, $p < 0.001$), highlighting the sensitivity of municipal effects to model specification and control inclusion. Financial-sector issuance is generally below one: *Financial Private Cumulative*$_{t-1}$ is consistently negative and statistically significant or near-significant across models (HR $\approx$ 0.622–0.753), consistent with a substitution pattern between financial-sector green bonds and non-financial corporate issuance.

The control variables underscore the central role of institutional capacity and development. *Liberal Democracy*$_{t-1}$ is strongly and positively associated with corporate issuance in Models 1–2 (HR = 16.6 and 42.4, both $p < 0.001$), indicating substantially higher adoption hazards in democracies. Treaty ratification is also positively associated with corporate adoption (HR = 2.03, $p < 0.05$ in Model 1; HR = 2.47, $p < 0.01$ in Model 2). Economic capacity measures point in the expected direction: GDP per capita and population are positive and significant in Models 1–2, and forest area is also positively associated with corporate issuance.

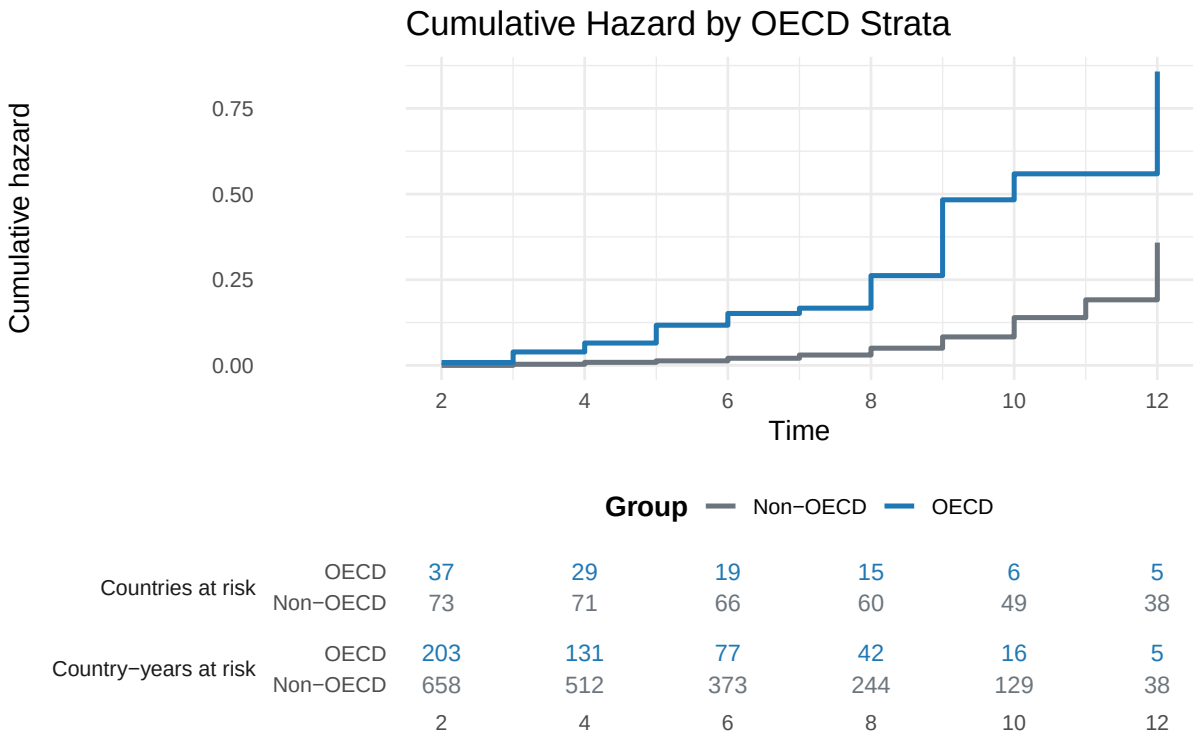
Overall, these reverse-causality models provide reassuring evidence for the main argument: there is no evidence that sovereign issuance increases the subsequent hazard of corporate issuance. If anything, the association runs in the opposite direction, although estimates are imprecise. This pattern strengthens the interpretation of the main results by suggesting that endogeneity concerns are more consistent with sovereigns responding to accumulated private-market infrastructure than with corporate markets being driven by sovereign "market leadership".

Table C.1: Cox Model Comparison

Variable	Model 1	Model 2	Model 3
Sovereign Cumulative$_{t-1}$	0.792	0.854	0.711
	(0.216)	(0.186)	(0.234)
Agency Cumulative$_{t-1}$	1.255	1.277†	1.434†
	(0.159)	(0.137)	(0.187)
Municipal Cumulative$_{t-1}$	0.873	0.827	1.794***
	(0.185)	(0.185)	(0.175)
Financial Private Cumulative$_{t-1}$	0.753†	0.622*	0.644†
	(0.169)	(0.195)	(0.241)
Financial Government Cumulative$_{t-1}$	0.826	0.932	1.345
	(0.328)	(0.2)	(0.278)
Liberal Democracy$_{t-1}$	16.555***	42.426***	
	(0.804)	(0.904)	
Climate Laws Total$_{t-1}$	1.001	1.018	
	(0.051)	(0.05)	
Climate Disasters Total$_{t-1}$	1.032	1.02	
	(0.03)	(0.028)	
Treaty Cumulative Ratified$_{t-1}$	0.984	0.985	
	(0.015)	(0.013)	
Treaty Ratified$_{t-1}$	2.025*	2.47**	
	(0.337)	(0.329)	
Log(Forest Area)$_{t-1}$	1.342**	1.251*	
	(0.09)	(0.091)	
Log(GDP per cap.)$_{t-1}$	2.729***	2.381**	
	(0.283)	(0.272)	
GDP Growth$_{t-1}$	1.047†	1.051†	
	(0.027)	(0.027)	
Log(Population)$_{t-1}$	1.845**	2.105***	
	(0.198)	(0.206)	
Financial Development$_{t-1}$	10.132†	25.27**	
	(1.237)	(1.245)	
Observations	728	760	861
Events	70	67	74
Countries	84	87	90
Concordance	0.848	0.863	0.633

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10.
Robust standard errors in parentheses.

Figure C.1



The cumulative hazard functions in Figure C.1 demonstrate the timing of corporate green bond issuance risk across OECD and non-OECD strata. OECD members exhibit substantially higher cumulative hazard throughout the observation period, with the hazard curve rising more steeply and reaching higher levels than their non-OECD counterparts. This pattern indicates that the instantaneous risk of first corporate green bond issuance is consistently and markedly higher among OECD states. The divergence between the two groups becomes particularly pronounced after time period 4, suggesting that OECD countries experience accelerated corporate adoption in later periods. The risk tables confirm this pattern, showing that OECD countries exit the risk set more rapidly, with fewer countries and country-years remaining at risk as time progresses.

The baseline survival curves in Figure C.2 present the probability that a country has not yet experienced corporate green bond issuance over time, conditional on the covariates included in Model 3. Both groups begin with survival probabilities near unity, reflecting that most countries remain at risk of corporate issuance in the initial periods. However, the OECD survival curve declines substantially more rapidly than the non-OECD curve, dropping to approximately 0.2 by the end of the observation period compared to roughly 0.6 for non-OECD countries. This steep decline indicates that OECD members experience earlier and more frequent corporate green bond adoption. The risk tables beneath the plot reveal that OECD countries systematically exit the risk set sooner, with only 5 countries and 5 country-years remaining at risk by the final time period, compared to 38 countries and 38 country-years for non-OECD states.

The proportional hazards assumption underlying the Cox regression models is assessed

Figure C.2

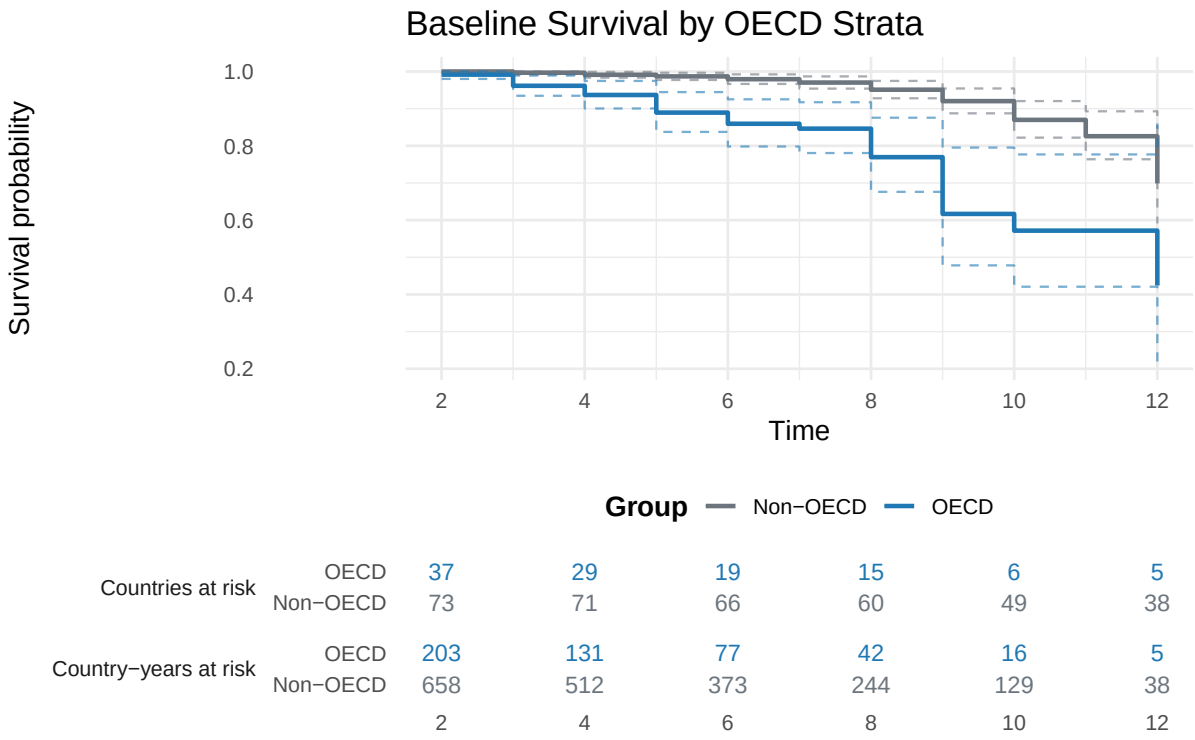


Table C.2: Proportional Hazards Assumption Test (Schoenfeld Residuals)

Variable	χ^2	p-value
Sovereign Cumulative$_{t-1}$	0.38	0.538
Agency Cumulative$_{t-1}$	3.73	0.054
Municipal Cumulative$_{t-1}$	0.02	0.889
Financial Private Cumulative$_{t-1}$	0.24	0.622
Financial Government Cumulative$_{t-1}$	0.01	0.934
Global test	9.70	0.084

using the Schoenfeld residuals test, with results presented in Table C.2. The test evaluates whether the hazard ratios remain constant over time, a critical assumption for valid Cox model interpretation. The global test statistic ($\chi^2 = 9.697$, p = 0.084) indicates that the proportional hazards assumption is not violated at conventional significance levels, supporting the validity of the Cox model specification.

Examining individual variables, most covariates satisfy the proportional hazards assumption with p-values well above 0.05. The agency cumulative variable shows the closest approach to violating this assumption (p = 0.054), suggesting some temporal variation in its effect on corporate green bond adoption hazard, though this remains marginally acceptable. All other bond market variables—sovereign, municipal, financial private, and financial gov-

ernment cumulative adoption—demonstrate stable hazard ratios over time (p-values ranging from 0.538 to 0.934). These results provide confidence that the estimated hazard ratios accurately represent the relative risk effects throughout the observation period, validating the use of standard Cox regression interpretation for the reverse causality analysis.

D Robustness to Explicit Bank Intermediation Controls

Table D.1 evaluates whether the infrastructure accumulation finding is robust to the inclusion of explicit measures of bank intermediation activity. If cumulative corporate bond counts capture the market infrastructure banks construct through repeated transactions, then the coefficient on *Corporate Private Cumulative* should remain stable when direct measures of bank presence are added to the model, and those direct measures should not independently predict sovereign adoption once the bond count is controlled for.

To test this mechanism, I construct two time-varying covariates measuring the cumulative stock of GBP consortium banks with prior in-country corporate green bond issuance experience: one counting all GBP consortium banks (*GBP Banks Corporate Exp.*$_{t-1}$) and one restricted to the four founding banks (JP Morgan, Citigroup, Bank of America, and Crédit Agricole) that drafted the Green Bond Principles (*GBP Founding Banks Corporate Exp.*$_{t-1}$).

Table D.1 reports four specifications: the baseline model and three augmented models adding the bank stock variables individually and jointly. The results support both predictions.

First, *Corporate Private Cumulative*$_{t-1}$ remains positive and statistically significant across all specifications, with hazard ratios ranging from 1.019 to 1.021. This stability indicates that the infrastructure accumulation finding is not sensitive to the inclusion of bank-level controls.

Second, neither bank stock variable is statistically significant in any specification. The estimated hazard ratios for GBP bank experience are close to unity, and the coefficients for founding-bank experience are imprecisely estimated. This pattern is consistent with the interpretation that bond counts already capture the channel through which banks operate: banks build market infrastructure through the transactions they underwrite, and the cumulative count of those transactions is therefore the appropriate measure of accumulated infrastructure.

The effect of *Corporate Private Cumulative.*$_{t-1}$ does not diminish when bank presence variables are added. This indicates that the bond count is not simply capturing whether banks operate in a country, but rather the infrastructure created through repeated transactions. In other words, what matters is not the presence of banks, but the cumulative market infrastructure built through corporate issuance.

Table D.1: Mechanism Test: Institutional Intermediation (Cox Proportional Hazards)

Variable	Baseline	All GBP Banks	Founding Banks	Both
Agency Cumulative$_{t-1}$	0.892**	0.893**	0.889**	0.897*
	(0.039)	(0.041)	(0.039)	(0.043)
Municipal Cumulative$_{t-1}$	1.007	1.007	1.007	1.007
	(0.004)	(0.004)	(0.004)	(0.004)
Financial Private Cumulative$_{t-1}$	0.991	0.99	0.994	0.981
	(0.033)	(0.036)	(0.033)	(0.035)
Financial Government Cumulative$_{t-1}$	0.994	0.993	0.998	0.99
	(0.021)	(0.023)	(0.022)	(0.025)
Corporate Private Cumulative$_{t-1}$	1.021**	1.021**	1.021**	1.019*
	(0.007)	(0.007)	(0.007)	(0.008)
Corporate Government Cumulative$_{t-1}$	1.016	1.017	1.014	1.018
	(0.013)	(0.014)	(0.014)	(0.015)
Electoral Power of Large Parties$_{t-1}$	1.218	1.215	1.22	1.176
	(0.156)	(0.158)	(0.159)	(0.161)
Climate Laws Total$_{t-1}$	1.003	1.002	1.005	0.994
	(0.036)	(0.036)	(0.035)	(0.034)
Climate Disasters Total$_{t-1}$	0.887	0.888	0.887	0.892
	(0.102)	(0.101)	(0.105)	(0.107)
Treaty Cumulative Ratified$_{t-1}$	0.948**	0.947*	0.948**	0.94**
	(0.019)	(0.022)	(0.019)	(0.021)
Treaty Ratified$_{t-1}$	2.455**	2.469**	2.444**	2.654**
	(0.306)	(0.311)	(0.309)	(0.313)
Log(Forest Area)$_{t-1}$	0.78*	0.783*	0.777*	0.805†
	(0.111)	(0.118)	(0.113)	(0.112)
Log(GDP per cap.)$_{t-1}$	2.418*	2.39*	2.603*	2.726*
	(0.357)	(0.382)	(0.412)	(0.405)
GDP Growth$_{t-1}$	0.974	0.974	0.979	0.984
	(0.024)	(0.026)	(0.028)	(0.03)
Log(Population)$_{t-1}$	2.382**	2.358*	2.481**	2.401**
	(0.297)	(0.333)	(0.333)	(0.326)
Financial Development$_{t-1}$	0.115	0.108	0.131	0.079
	(1.686)	(1.754)	(1.638)	(1.804)
GBP Banks Corporate Exp.$_{t-1}$		1.007		1.129
		(0.062)		(0.118)
GBP Founding Banks Corporate Exp.$_{t-1}$			0.912	0.649
			(0.184)	(0.38)
Observations	681	681	681	681
Events	34	34	34	34
Countries	70	70	70	70
Concordance	0.757	0.759	0.761	0.761

Note: Hazard ratios reported. Robust standard errors clustered by country in parentheses. All models stratified by OECD membership. GBP Banks Corporate Exp.$_{t-1}$ counts distinct GBP consortium banks with prior corporate green bond issuance experience in the country. GBP Founding Banks Corporate Exp.$_{t-1}$ restricts this to the four founding banks. *** p<0.001, ** p<0.01, * p<0.05, † p<0.10.

E Robustness Check: Value-Weighted Cumulative Issuance

Table E.1 examines whether the infrastructure accumulation effect is driven by the total value of prior corporate green bond issuance rather than the number of transactions. If the mechanism operates through deal volume—such that larger issuances generate more market infrastructure—then value-weighted measures should outperform or displace bond counts. Value-weighted alternatives are constructed by taking the natural log of cumulative issuance in millions of USD (plus one to accommodate zeros) and standardizing to z-scores, so that hazard ratios correspond to a one-standard-deviation increase in log cumulative issuance value.

Table E.1 reports three specifications: the count-based baseline, a value-weighted specification, and a horse-race model including both simultaneously. The results consistently favor the count-based measure. In the value-weighted specification (Column 2), the value-weighted corporate private issuance measure is not statistically significant, while the count-based estimate remains positive and statistically significant in the baseline and combined models (HR $= 1.018$, $p < 0.01$ in both cases). In the horse-race specification, the count variable remains significant while the value-weighted alternative remains statistically indistinguishable from zero, indicating that incorporating issuance values does not improve predictive power.

The agency substitution effect is similarly robust. *Agency Cumulative*$_{t-1}$ is negative and statistically significant in both the count and combined specifications (HR ≈ 0.90, $p < 0.01$), while the value-weighted agency measure is not statistically significant.

These results are theoretically informative. The infrastructure accumulation mechanism predicts that each corporate transaction contributes to the stock of procedural templates, verified issuers, trained intermediaries, and investor familiarity that reduces sovereign entry costs. Under this account, what matters is the number of transactions—each adding marginally to accumulated market infrastructure—rather than the dollar volume they represent. A \$50 million deal and a \$500 million deal generate comparable infrastructure benefits. The dominance of bond counts over issuance values in the horse-race specification is therefore consistent with a transaction-based interpretation of the mechanism.

Table E.1: Robustness Check: Value-Weighted Cumulative Issuance (Cox Proportional Hazards)

Variable	Count	Value-Weighted	Both
Corporate Private Cumulative$_{t-1}$	1.018**		1.018**
	(0.006)		(0.006)
Corporate Government Cumulative$_{t-1}$	1.014		1.014
	(0.013)		(0.014)
Agency Cumulative$_{t-1}$	0.9**		0.9**
	(0.037)		(0.037)
Municipal Cumulative$_{t-1}$	1.006		1.006
	(0.005)		(0.005)
Financial Private Cumulative$_{t-1}$	0.986		0.986
	(0.034)		(0.034)
Financial Government Cumulative$_{t-1}$	0.997		0.997
	(0.021)		(0.021)
Electoral Power of Large Parties$_{t-1}$	1.208	1.15	1.209
	(0.151)	(0.145)	(0.153)
Climate Laws Total$_{t-1}$	1.001	1.028	1.002
	(0.035)	(0.04)	(0.038)
Climate Disasters Total$_{t-1}$	0.876	0.918	0.875
	(0.101)	(0.095)	(0.101)
Treaty Cumulative Ratified$_{t-1}$	0.949**	0.966*	0.949*
	(0.019)	(0.017)	(0.021)
Treaty Ratified$_{t-1}$	2.478**	2.604*	2.468**
	(0.304)	(0.385)	(0.312)
Log(Forest Area)$_{t-1}$	0.782*	0.813	0.781*
	(0.107)	(0.129)	(0.109)
Log(GDP per cap.)$_{t-1}$	1.741*	1.972**	1.74*
	(0.234)	(0.211)	(0.232)
GDP Growth$_{t-1}$	0.98	0.979	0.98
	(0.024)	(0.023)	(0.024)
Log(Population)$_{t-1}$	2.104**	1.974*	2.107**
	(0.249)	(0.269)	(0.257)
Corporate Private Cumulative, Log($M)$_{t-1}$		0.994	0.98
		(0.333)	(0.338)
Corporate Government Cumulative, Log($M)$_{t-1}$		1.392*	
		(0.156)	
Agency Cumulative, Log($M)$_{t-1}$		0.681	
		(0.502)	
Municipal Cumulative, Log($M)$_{t-1}$		1.467	
		(0.397)	
Financial Private Cumulative, Log($M)$_{t-1}$		0.711†	
		(0.204)	
Financial Government Cumulative, Log($M)$_{t-1}$		0.612*	
		(0.199)	
Observations	765	765	765
Events	37	37	37
Countries	78	78	79
Concordance	0.725	0.737	0.725

Note: Hazard ratios reported. Robust standard errors clustered by country in parentheses. All models stratified by OECD membership. Value-weighted covariates are standardized (z-scores); hazard ratios correspond to a one-standard-deviation increase in log cumulative issuance value (in $M, plus one to accommodate zeros). *** p<0.001, ** p<0.01, * p<0.05, † p<0.10.

F Robustness Check: Excluding Political Signaling Countries

Table F.1 re-estimates the main Cox proportional hazards models after excluding the ten countries identified as sovereign green bond adopters with no prior green bond ecosystem of any kind—no corporate, agency, municipal, or other non-sovereign issuance preceding their sovereign entry. As discussed in Section 2.2, these countries (Andorra, Bahamas, Belize, Benin, Fiji, Gabon, Honduras, North Macedonia, Serbia, and Seychelles) appear to have adopted sovereign green bonds through political signaling channels, often in conjunction with international climate commitments, rather than in response to accumulated market infrastructure. Including them in the main estimation sample is conservative—their adoption follows a pathway orthogonal to the bank-mediated infrastructure accumulation mechanism—but their presence could attenuate estimated infrastructure effects if adoption through signaling is not predicted by market development variables.

The results are substantively unchanged. *Corporate Private Cumulative*$_{t-1}$ remains positive and statistically significant across all three specifications, with hazard ratios of 1.022 ($p < 0.01$), 1.017 ($p < 0.05$), and 1.016 ($p < 0.05$), closely mirroring the main results. The agency substitution effect similarly persists: *Agency Cumulative*$_{t-1}$ is negative across all models, reaching conventional significance in Model 1 (HR = 0.881, $p < 0.01$) and remaining marginally significant in Models 2 and 3.

The stability of both key coefficients across the full and restricted samples indicates that the infrastructure accumulation and agency substitution findings are not driven by the political signaling cases but instead reflect the broader market-building dynamics identified by the theory.

Table F.1: Robustness Check: Excluding Political Signaling Countries

Variable	Model 1	Model 2	Model 3
Agency Cumulative$_{t-1}$	0.881**	0.935†	0.93†
	(0.041)	(0.036)	(0.037)
Municipal Cumulative$_{t-1}$	1.007†	1.002	1.002
	(0.004)	(0.003)	(0.004)
Financial Private Cumulative$_{t-1}$	0.988	0.978	0.981
	(0.036)	(0.031)	(0.03)
Financial Government Cumulative$_{t-1}$	0.991	0.988	0.998
	(0.02)	(0.016)	(0.015)
Corporate Private Cumulative$_{t-1}$	1.022**	1.017*	1.016*
	(0.008)	(0.007)	(0.008)
Corporate Government Cumulative$_{t-1}$	1.021†	1.013	1.009
	(0.012)	(0.009)	(0.01)
Climate Laws Total$_{t-1}$	0.977	0.963	0.953
	(0.035)	(0.035)	(0.036)
Climate Disasters Total$_{t-1}$	0.859	1	0.978
	(0.102)	(0.086)	(0.086)
Treaty Cumulative Ratified$_{t-1}$	0.948**	1	0.993
	(0.02)	(0.015)	(0.016)
Treaty Ratified$_{t-1}$	2.492**	2.018*	2.16*
	(0.327)	(0.35)	(0.33)
Log(Forest Area)$_{t-1}$	0.729**	0.832	0.856
	(0.12)	(0.114)	(0.095)
Log(GDP per cap.)$_{t-1}$	3.981***	3.898***	2.704**
	(0.399)	(0.326)	(0.338)
GDP Growth$_{t-1}$	0.963	0.971	0.973
	(0.028)	(0.024)	(0.025)
Log(Population)$_{t-1}$	3.815***	2.55***	2.477***
	(0.261)	(0.238)	(0.246)
Financial Development$_{t-1}$	0.041†	0.048†	0.097
	(1.902)	(1.62)	(1.659)
Electoral Power of Large Parties$_{t-1}$	1.372†		
	(0.165)		
Electoral Local Power$_{t-1}$		1.072	
		(0.206)	
Liberal Democracy$_{t-1}$			7.761*
			(1.005)
Observations	740	883	924
Events	34	41	43
Countries	75	90	93
Concordance	0.823	0.763	0.775

G Robustness Check: Excluding the United States

The United States represents an extreme case in the agency substitution analysis: with 9,379 agency green bonds issued since 2012—more than any other country by a substantial margin—and no sovereign green bond issuance through the end of the observation period, the U.S. could exert disproportionate influence on the agency substitution coefficient. Table G.1 re-estimates the main Cox proportional hazards models after excluding the United States from the estimation sample.

The results are substantively unchanged. *Corporate Private Cumulative*$_{t-1}$ remains positive and statistically significant across all three specifications, with hazard ratios of 1.021 ($p < 0.01$), 1.017 ($p < 0.05$), and 1.017 ($p < 0.05$), closely mirroring the main results. The agency substitution effect similarly persists: *Agency Cumulative*$_{t-1}$ is negative across all three models, reaching conventional significance in Model 1 (HR = 0.89, $p < 0.01$) and remaining statistically significant in Models 2 and 3 (HR = 0.935 and 0.932).

All other substantive findings remain stable. Treaty ratification continues to increase the likelihood of sovereign issuance, and GDP per capita and population remain positive predictors of adoption. These results indicate that the infrastructure accumulation and agency substitution findings are not driven by the United States' outsized agency green bond activity but instead reflect broader market dynamics.

Table G.1: Robustness Check: Excluding the United States (Cox Prop. Hazards)

Variable	Model 1	Model 2	Model 3
Agency Cumulative$_{t-1}$	0.89**	0.935*	0.932*
	(0.038)	(0.034)	(0.035)
Municipal Cumulative$_{t-1}$	1.016	1.016	1.013
	(0.018)	(0.017)	(0.017)
Financial Private Cumulative$_{t-1}$	0.989	0.983	0.984
	(0.035)	(0.029)	(0.028)
Financial Government Cumulative$_{t-1}$	0.998	0.998	1.004
	(0.019)	(0.014)	(0.014)
Corporate Private Cumulative$_{t-1}$	1.021**	1.017*	1.017*
	(0.007)	(0.007)	(0.007)
Corporate Government Cumulative$_{t-1}$	1.013	1.005	1.003
	(0.012)	(0.01)	(0.01)
Climate Laws Total$_{t-1}$	1.001	0.977	0.971
	(0.037)	(0.037)	(0.037)
Climate Disasters Total$_{t-1}$	0.893	1.02	1.003
	(0.102)	(0.079)	(0.081)
Treaty Cumulative Ratified$_{t-1}$	0.948**	0.997	0.994
	(0.019)	(0.014)	(0.015)
Treaty Ratified$_{t-1}$	2.528**	2.136*	2.181*
	(0.314)	(0.326)	(0.317)
Log(Forest Area)$_{t-1}$	0.769*	0.866	0.906
	(0.116)	(0.101)	(0.096)
Log(GDP per cap.)$_{t-1}$	2.455*	2.634**	1.994*
	(0.353)	(0.308)	(0.292)
GDP Growth$_{t-1}$	0.974	0.978	0.982
	(0.024)	(0.021)	(0.022)
Log(Population)$_{t-1}$	2.411**	1.714*	1.629†
	(0.296)	(0.257)	(0.263)
Financial Development$_{t-1}$	0.101	0.103	0.179
	(1.68)	(1.438)	(1.448)
Electoral Power of Large Parties$_{t-1}$	1.221		
	(0.156)		
Electoral Local Power$_{t-1}$		1.004	
		(0.189)	
Liberal Democracy$_{t-1}$			2.993
			(1.018)
Observations	765	923	964
Events	37	46	48
Countries	78	95	98
Concordance	0.731	0.665	0.662

Note: *** p<0.001, ** p<0.01, * p<0.05, † p<0.10. Hazard ratios reported. Robust standard errors clustered by country in parentheses. All models stratified by OECD membership. The United States is excluded given its outsized agency green bond activity (9,000+ bonds) and no sovereign green bond issuance, which could disproportionately influence the agency substitution coefficient.

H Additional Evidence on Bank Motivations

H.1 The "Financed, Not Funded" Philosophy

The bank-led diffusion of green bonds reflected a particular ideological orientation toward climate finance. Michael Eckhart, whose career spanned renewable energy project finance and sustainable banking, articulated a sharp distinction between government "funding" and private "financing":

> "The Department of Energy R&D budget doesn't have enough money to pay for all the $100 billion a year of PV we install every year today. Where'd that money come from? It came from Wall Street... We wanted to do something totally different, which is instead of lobbying Congress, we would set out to lobby Wall Street because we realized... that's where the money is. There's no money in Washington" (Eckhart, 2021).

This philosophy—that climate solutions required mobilizing private capital markets rather than expanding government expenditure—shaped the GBP's design and the banks' subsequent market-building efforts. The principles deliberately positioned green bonds as a market-based mechanism compatible with existing debt capital market practices, requiring no regulatory mandate or public subsidy. By framing green finance as an investment opportunity rather than a policy intervention, banks could approach corporate clients as offering a competitive product rather than requesting adherence to external standards.

H.2 Additional Statements from GBP Co-Authors

The founding banks were explicit about their motivations for creating market standards. Marilyn Ceci of JPMorgan Chase emphasized the importance of debt capital markets expertise:

> "I think the success of the development of the GBP is that you really had four bankers working on this, not the team in sustainability, because the thing you know about the bond market as a general rule is this: it needs to be simple, it needs to be liquid, and it needs to be big" (Ceci, 2024).

Ceci was also explicit about learning from prior market failures:

> "We had watched the CDO market and the CDOs squared—we'd all seen many different markets rise and fall. And so with that you say, wait a minute, if we really like this market and we really think it has legs... we want to make sure we publish some guidance" (Ceci, 2024).

The co-authors acknowledged tension with environmental advocates who preferred stricter standards. Ceci noted that they "met with quite a lot of pushback from some of the environmental community who basically said you're not going far enough," but emphasized that "it was absolutely critical to make sure and remind ourselves consistently that perfect was the enemy of good" (Ceci, 2024).

Suzanne Buchta of Bank of America Merrill Lynch emphasized the accessibility of the framework:

> "The brilliant thing about the concept is that it's so simple and so easily accessible. A green bond would offer investors a clear, verifiable connection between their financial commitment and a project that helps the climate" (Heimer, 2018).

At the GBP launch, Buchta stated the banks' explicit goal: "In co-authoring these principles we attempt to help standardize the product and we hope to catalyze investment into environmentally sustainable projects" (Bank of America Merrill Lynch et al., 2014). Tanguy Claquin of Crédit Agricole CIB similarly noted that the principles represented "an important first step towards a more coherent approach to the market of Green and Sustainability Bonds, which will ultimately increase its attractiveness for investors" (Bank of America Merrill Lynch et al., 2014).

H.3 Reputational Risk Management

Writing a year after the GBP launch, Claquin acknowledged the reputational stakes that motivated bank investment in market governance:

> "The current green bonds market, which is self-regulated, is exposed to a major risk, namely what would happen if an issuer blatantly violated its 'green' commitments?"

The market managed this risk, he observed, because "most actors (issuers, investors and banks) have recognised the best practices drawn up by four banks" (Claquin, 2015). This self-regulatory framing—emphasizing voluntary coordination among market participants rather than external oversight—was central to the banks' market-building strategy.

I Climate Treaty Amplification

Treaty ratification emerges as one of the strongest predictors of sovereign green bond adoption. *Treaty Ratified*$_{t-1}$ produces hazard ratios ranging from 2.18 to 2.96 across specifications, indicating that ratifying a climate treaty more than doubles—and in some models nearly triples—the instantaneous likelihood of sovereign issuance. Unlike the cumulative issuance variables, this effect does not compound over time; it captures the immediate impact of formal international commitment on sovereign financing decisions.

This finding supports the theoretical prediction that climate commitments amplify the effect of market infrastructure. International treaties create political pressure for visible climate action, but governments face uncertainty about how to translate commitments into concrete policy. Accumulated market infrastructure—built through prior corporate issuance—provides a ready mechanism for demonstrating climate commitment through sovereign green bond programs. Treaties and infrastructure are thus complementary: treaties increase demand for sovereign climate signaling, while infrastructure reduces the costs and risks of market entry.

The treaty effect also helps explain the temporal pattern documented in Figure 2. The median four-year lag between first corporate and first sovereign issuance reflects the time required for market infrastructure to accumulate and for political conditions—often catalyzed by treaty commitments—to favor sovereign adoption. Countries do not respond immediately to corporate activity because infrastructure must reach sufficient depth, and political impetus must develop, before sovereign programs become attractive.